
EMCH 574 – Vibration and Waves

Homework 01

Student J.C. Vaught

Due date Sep. 16, 2026

Contents

1	Pendulum	3
1.1	A.1 Analyze Pendulum	3
1.2	A.2 Calculate Pendulum Response to Initial Displacement	8
1.3	A.3 Calculate Pendulum Response to Impact	12
2	Spring-Mass-Damper Vibrating Systems	17
2.1	B.1 Analyze Spring-Mass Vibrating System	17
2.2	B.2 Analyze Spring-Mass-Damper Vibrating System	20
2.3	B.3 Calculate Spring-Mass-Damper Response to Initial Displacement	27
2.4	B.4 Calculate Truck Suspension Response	33
2.5	B.5 Overdamped Spring-Mass-Damper Systems	43
2.6	B.6 Analyze Spring-Mass-Damper Forced Vibration	50
2.7	B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation	58
2.8	B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact	67
3	Extra Credit Challenge Problems	74
3.1	C.1 Analyze Vertical Spring-Mass Vibration System	74
3.2	C.2 Analyze Vertical Flex-Beam Vibration	78
3.3	C.3 Analyze Horizontal Flex-Beam Vibration	83
3.4	C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement	87
3.5	C.5 Analyze Spring-Mass Forced Vibration	91
3.6	C.6 Calculate Cantilever Response to Suddenly Applied Weight	95
3.7	C.7 Analyze Flex-Beam Pendulum	100
3.8	C.8 Calculate Flex-Beam Pendulum Response to Impact	105
3.9	C.9 Complex Exponentials and Trigonometric Functions	110

Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#).
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) [reference](#).
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the *y-axis* scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Pendulum

A.1 Analyze Pendulum

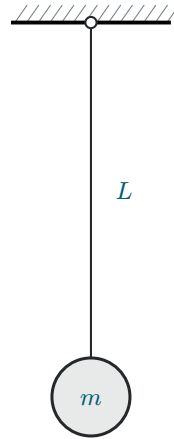


Figure 1: Pendulum of length L and mass m .

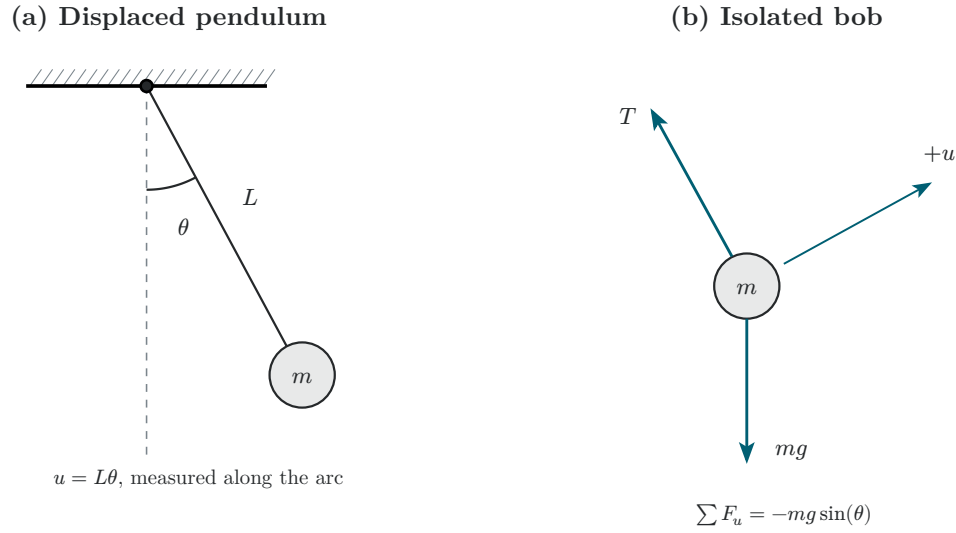
Consider a pendulum of length L and mass m as shown in Figure 1.

Do the following.

- (1) Draw the pendulum displaced by an angle θ from the vertical.
- (2) Draw free-body diagram (FBD).
- (3) Write Newton's law of motion (NLM) equation.
- (4) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (5) Deduce the governing equation.
- (6) Assume $\theta(t) = e^{st}$ and substitute into the governing equation.
- (7) Deduce and solve the characteristic equation.
- (8) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (9) Define the natural frequency ω_n .
- (10) Express the solutions s_1 and s_2 of the characteristic equation in terms of ω_n .
- (11) Write the solution in terms of exponential functions of $\omega_n t$.
- (12) Write the solution in terms of trigonometric functions of $\omega_n t$.
- (13) Write the Hz frequency f_n .

- (14) Write the period of oscillation τ .
- (15) Write the particle motion $u(t)$ in terms of:
- (a) Exponential functions.
 - (b) Trigonometric functions.
- (16) Solve the initial value problem (IVP) and find the response $u(t)$ for:
- (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to A.1

**Figure 2:** Free-body diagram for Problem A.1.

I take positive θ in the direction shown and define $u = L\theta$ as the tangential arc displacement, which gives $\ddot{u} = L\ddot{\theta}$. I assume a point mass, a massless inextensible rod or taut string of length L , a fixed frictionless pivot, uniform gravity, planar motion, and negligible air resistance.

The weight is mg , and the tension T has no tangential component. Therefore, the tangential force is the restoring component of gravity,

$$\sum F_u = -mg \sin \theta = m\ddot{u} = mL\ddot{\theta} \quad (\text{A.1.3})$$

For small angular displacement in radians, $\sin \theta \approx \theta$. Substituting this approximation gives the equation of motion

$$mL\ddot{\theta} + mg\theta = 0 \quad (\text{A.1.4})$$

Dividing by mL gives the governing equation

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \quad (\text{A.1.5})$$

Since the governing equation is linear with constant coefficients, the trial response $\theta = e^{st}$ applies. Then $\ddot{\theta} = s^2 e^{st}$, so substitution gives

$$\left(s^2 + \frac{g}{L}\right) e^{st} = 0.$$

Because $e^{st} \neq 0$, the characteristic equation is

$$\boxed{s^2 + \frac{g}{L} = 0} \quad (\text{A.1.7})$$

Since $g/L > 0$, solving this equation gives purely imaginary roots. Therefore, the answer to item 8 is

$$\boxed{\text{Imaginary}} \quad (\text{A.1.8.b})$$

Defining the natural angular frequency by $\omega_n^2 = g/L$ gives

$$\boxed{\omega_n = \sqrt{\frac{g}{L}}} \quad (\text{A.1.9})$$

Using this definition, the roots become

$$\boxed{s_1 = i\omega_n, \quad s_2 = -i\omega_n} \quad (\text{A.1.10})$$

The two roots produce the exponential angular response

$$\boxed{\theta(t) = D_1 e^{i\omega_n t} + D_2 e^{-i\omega_n t}} \quad (\text{A.1.11})$$

Using Euler's identity, the same response can be written as

$$\boxed{\theta(t) = A_\theta \cos(\omega_n t) + B_\theta \sin(\omega_n t)} \quad (\text{A.1.12})$$

Because $\omega_n = 2\pi f_n$ and $f_n = 1/\tau$, the frequency and period are

$$\boxed{f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}} \quad (\text{A.1.13})$$

$$\boxed{\tau = \frac{1}{f_n} = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{L}{g}}} \quad (\text{A.1.14})$$

Multiplying the angular response by L gives the particle-motion response. In exponential form, this is

$$\boxed{u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t}} \quad (\text{A.1.15.a})$$

Euler's identity also gives the trigonometric forms

$$\boxed{u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) = C \sin(\omega_n t + \varphi)} \quad (\text{A.1.15.b})$$

For real A and B , the amplitude and phase are $C = \sqrt{A^2 + B^2}$ and $\varphi = \text{atan2}(A, B)$.

Differentiating the trigonometric response and evaluating it at $t = 0$ gives

$$u(0) = A = u_0, \quad \dot{u}(0) = \omega_n B = v_0, \quad v_0 = \dot{u}_0.$$

With initial displacement only, $v_0 = 0$, so

$$\boxed{u(t) = u_0 \cos(\omega_n t)} \quad (\text{A.1.16.i})$$

With initial velocity only, $u_0 = 0$, so

$$\boxed{u(t) = \frac{v_0}{\omega_n} \sin(\omega_n t)} \quad (\text{A.1.16.ii})$$

With both initial conditions applied, the response is

$$\boxed{u(t) = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t)} \quad (\text{A.1.16.iii})$$

The corresponding exponential constants are

$$C_1 = \frac{1}{2} \left(u_0 - \frac{iv_0}{\omega_n} \right), \quad C_2 = \frac{1}{2} \left(u_0 + \frac{iv_0}{\omega_n} \right).$$

The mass cancels when the governing equation is divided by m , so the ideal small-angle pendulum frequency depends on g and L , but not on m .

A.2 Calculate Pendulum Response to Initial Displacement

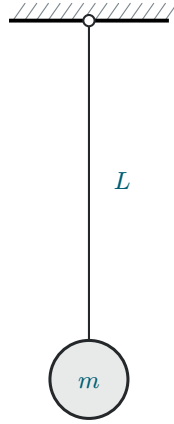


Figure 3: Pendulum of length L and mass m .

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}} = 565 \text{ mm}$ and mass $m = 52 \text{ kg}$ as shown in Figure 3. When manufactured in series production, the pendulum lengths are not exact because of manufacturing tolerances. The lengths of the ten pendulum sets are

$$L = [561.535, 562.691, 563.845, 563.845, 565.000, \\ 565.000, 566.155, 566.155, 567.309, 568.464] \text{ mm}.$$

Do the following.

- (1) For the ten pendulum sets, calculate.
 - (a) Natural frequency ω_n in rad/s.
 - (b) Hz natural frequency f_n .
 - (c) Period of oscillation τ .
 - (d) Display in table format L, ω_n, f_n, τ .
- (2) Mean values: calculate and display the mean values of ω_n, f_n, τ .
- (3) Initial value problem (IVP). Assume initial displacement $u_0 = 2 \text{ mm}$ and zero initial velocity. Use the mean values of ω_n, f_n , and τ to complete items (a)–(c).
 - (a) Calculate and display the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
 - (b) Calculate and display the duration $t_{N_{\text{osc}}}$ of N_{osc} oscillations and the corresponding position of the mass $u_{N_{\text{osc}}}$ for $N_{\text{osc}} = 10$.
 - (c) Plot the response of the system for N_{osc} oscillations and use a data tip to find $u(t_1)$ at t_1 .

Solution to A.2

Since the pendulum relation uses SI units, each length is first converted from millimetres to metres. For set j , the relation from A.1 then gives

$$\omega_{n,j} = \sqrt{\frac{g}{L_j}} \quad (\text{A.2.1.a})$$

The frequency and period follow from $\omega_{n,j} = 2\pi f_{n,j}$ and $f_{n,j} = 1/\tau_j$, so

$$f_{n,j} = \frac{\omega_{n,j}}{2\pi} \quad (\text{A.2.1.b})$$

$$\tau_j = \frac{2\pi}{\omega_{n,j}} \quad (\text{A.2.1.c})$$

Using $g = 9.81 \text{ m/s}^2$ and the ten lengths gives the following values for A.2.1.d.

Set	L (mm)	ω_n (rad/s)	f_n (Hz)	τ (s)
1	561.535	4.18	0.665	1.503
2	562.691	4.18	0.665	1.505
3	563.845	4.17	0.664	1.506
4	563.845	4.17	0.664	1.506
5	565.000	4.17	0.663	1.508
6	565.000	4.17	0.663	1.508
7	566.155	4.16	0.663	1.509
8	566.155	4.16	0.663	1.509
9	567.309	4.16	0.662	1.511
10	568.464	4.15	0.661	1.513

The mean of each calculated column follows from $\bar{q} = \frac{1}{10} \sum_{j=1}^{10} q_j$. Therefore, the values required in A.2.2 are

$$\overline{\omega_n} = 4.17 \text{ rad/s} \quad (\text{A.2.2})$$

$$\overline{f_n} = 0.663 \text{ Hz} \quad (\text{A.2.2})$$

$$\overline{\tau} = 1.508 \text{ s} \quad (\text{A.2.2})$$

For A.2.3, the initial velocity is zero, so the A.1 response reduces to

$$u(t) = u_0 \cos(\overline{\omega_n} t) = 2 \cos(4.166893t) \text{ mm} \quad (\text{A.2.3})$$

At $t_1 = 10$ s, substitution into the response gives

$$u(t_1) = 2 \cos(4.166893 \times 10) \text{ mm} = -1.352 \text{ mm} \quad (\text{A.2.3.a})$$

For ten oscillations, multiply the mean period by $N_{\text{osc}} = 10$. The duration and corresponding position are

$$t_{N_{\text{osc}}} = 10\bar{\tau} = 15.08 \text{ s} \quad (\text{A.2.3.b})$$

$$u_{N_{\text{osc}}} = u(10\bar{\tau}) = 2.000 \text{ mm} \quad (\text{A.2.3.b})$$

Using the boxed A.2.3 expression over $0 \leq t \leq 10\bar{\tau}$ gives the ten-oscillation response below. Since the marker is placed at $t_1 = 10$ s, the plotted value is

$$u(10) = -1.352 \text{ mm} \quad (\text{A.2.3.c})$$

A2. Pendulum released from 2 mm

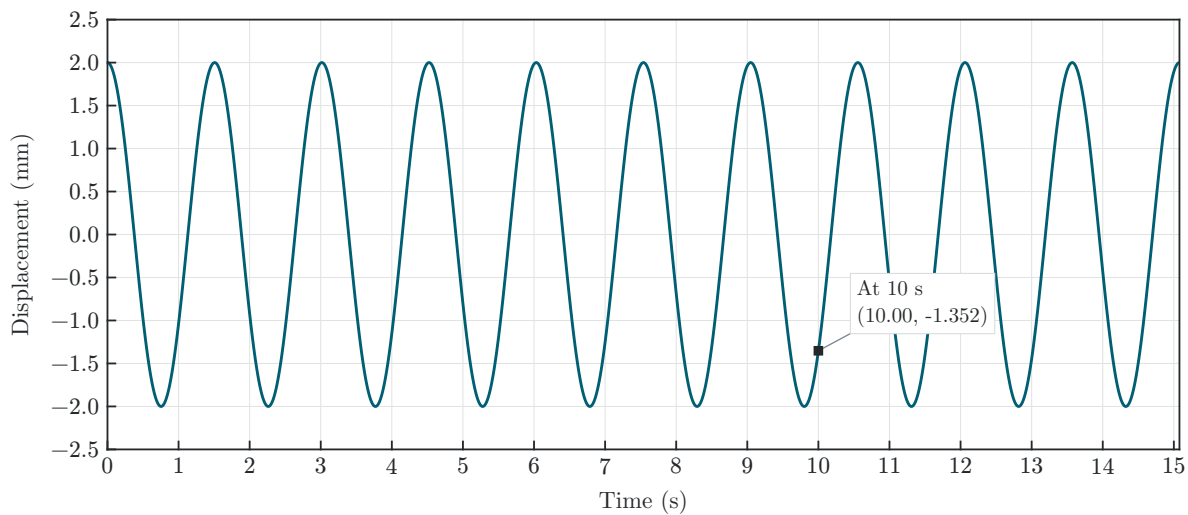


Figure 4: A.2.3.c. Pendulum response over ten oscillations with a data tip at $t = 10$ s.

MATLAB code. *Script used to reproduce Figure 4.*

```
01 u0 = 2;  
02 wn = 4.166893;  
03 t1 = 10;  
04 Nosc = 10;  
05  
06 tau = 2*pi/wn;  
07 t = linspace(0,Nosc*tau,1000);  
08 u = u0*cos(wn*t);  
09  
10 plot(t,u,t1,u0*cos(wn*t1),'ks');  
11 xlabel('Time (s)');  
12 ylabel('u (mm)');
```

A.3 Calculate Pendulum Response to Impact

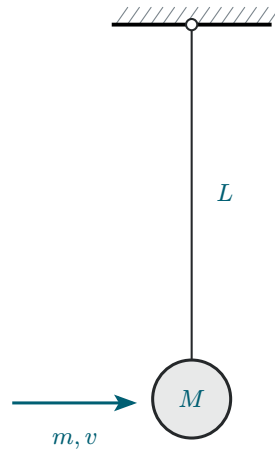


Figure 5: *Impact excitation of a pendulum system.*

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L = 565 \text{ mm}$ and mass $M = 52 \text{ kg}$ as shown in Figure 5. The pendulum is initially at rest hanging vertically. A small projectile of mass $m = 22 \text{ g}$ and velocity $v = 945 \text{ m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it.
- (3) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (4) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (5) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (4).
- (6) Discuss your results.

Solution to A.3

The input values required for A.3.1 are shown below. Since the response equations use SI units, the listed values are converted before substitution. The projectile mass is denoted by m_p so that it remains distinct from the pendulum mass M .

Quantity	Value	Unit
g	9.81	m/s ²
L	0.565	m
M	52.0	kg
m_p	0.0220	kg
v_p	945	m/s
t_1	10.0	s
N_{osc}	10	1

For the pendulum, the small-angle restoring motion gives the undamped equation

$$\ddot{u} + \frac{g}{L}u = 0.$$

Therefore, the natural frequency does not depend on the total mass, because the mass cancels from the ideal pendulum equation. The natural frequency and period are

$$\omega_n = \sqrt{\frac{g}{L}} = 4.16687 \text{ rad/s} \quad (\text{A.3.2})$$

$$\tau = \frac{2\pi}{\omega_n} = 1.50789 \text{ s} \quad (\text{A.3.2})$$

During the short impact, the projectile embeds in the pendulum mass, so linear momentum is conserved. Since the pendulum mass starts at rest, the common velocity immediately after impact satisfies

$$m_p v_p + M(0) = (M + m_p) \dot{u}_0.$$

Solving for the initial velocity gives

$$\dot{u}_0 = \frac{m_p v_p}{M + m_p} = 0.399639 \text{ m/s} \quad (\text{A.3.3})$$

The pendulum is still vertical immediately after impact, so $u_0 = 0$. Substitution into the undamped initial-value solution gives

$$u(t) = u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t).$$

Because $u_0 = 0$, the response becomes

$$u(t) = \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) = 95.9085 \sin(4.16687t) \text{ mm} \quad (\text{A.3.4})$$

At $t_1 = 10$ s, substitution into the response gives

$$u(t_1) = 95.9085 \sin(4.16687 \times 10) \text{ mm} = -70.6461 \text{ mm} \quad (\text{A.3.4})$$

Ten oscillations span the interval

$$t_{\text{end}} = N_{\text{osc}} \tau = 10(1.50789) = 15.0789 \text{ s} \quad (\text{A.3.5})$$

The response reaches its first positive maximum one quarter period after impact. The largest absolute response is therefore

$$|u|_{\text{max}} = \frac{\dot{u}_0}{\omega_n} = 95.9085 \text{ mm} \quad (\text{A.3.5.a})$$

The ten-oscillation plot includes data tips at the first maximum and at $t_1 = 10$ s. The plotted readings are

$$u_{\text{max}} = 95.9 \text{ mm} \quad (\text{A.3.5.a})$$

$$u(10 \text{ s}) = -70.6 \text{ mm} \quad (\text{A.3.5.b})$$

A3. Response after an embedded-projectile impact

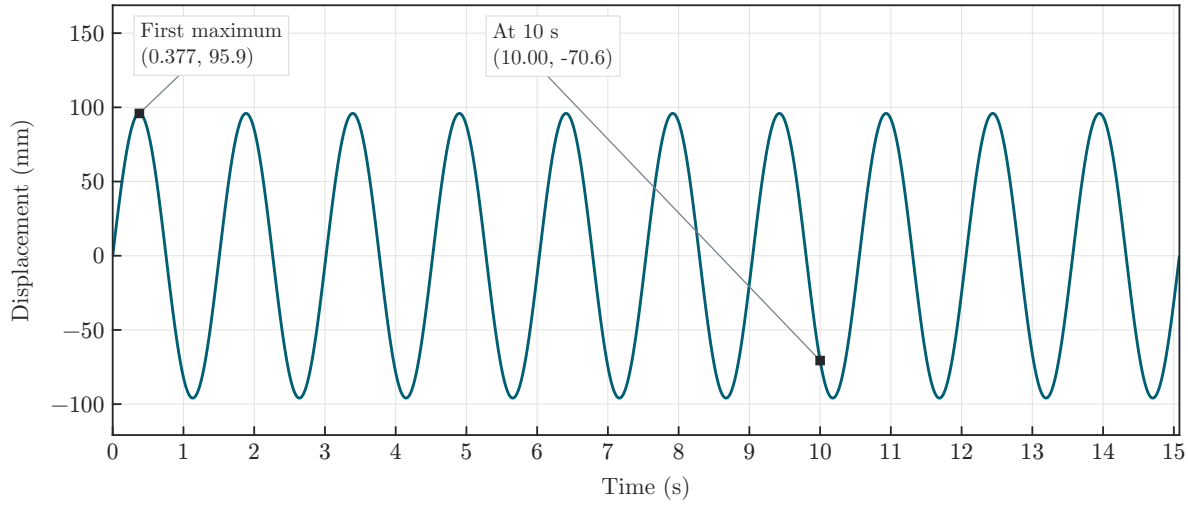


Figure 6: A.3.5.a-b. Impact response with data tips.

The projectile transfers its momentum to the combined mass, which produces a nonzero initial velocity but no initial displacement. The response consequently starts at zero, reaches its first maximum at $t = \tau/4 = 0.377$ s, and then repeats with constant amplitude because damping is neglected. The maximum displacement corresponds to an angular amplitude of approximately $u_{\max}/L = 0.170$ rad = 9.73° , so the small-angle model is reasonable for this calculation. At 10 s, the pendulum is moving through the negative side of the motion rather than being at a maximum, which gives the plotted value of -70.6 mm.

MATLAB code. *Script used to plot the ten-oscillation response.*

```
01 g = 9.81;
02 L = 0.565;
03 M = 52;
04 mp = 0.022;
05 vp = 945;
06 t1 = 10;
07 Nosc = 10;
08 u0 = 0;
09
10 wn = sqrt(g/L);
11 tau = 2*pi/wn;
12 udot0 = mp*vp/(M+mp);
13 A = udot0/wn;
14
15 t = linspace(0,Nosc*tau,1000);
16 u = u0*cos(wn*t) + A*sin(wn*t);
17 u1 = u0*cos(wn*t1) + A*sin(wn*t1);
18
19 plot(t,1000*u,tau/4,1000*A,'ks',t1,1000*u1,'ks');
20 xlabel('Time (s)');
21 ylabel('u (mm)');
22 grid on;
```

2 Spring-Mass-Damper Vibrating Systems

B.1 Analyze Spring-Mass Vibrating System

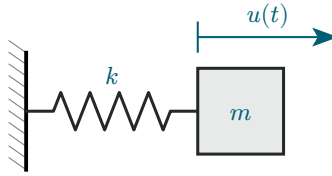


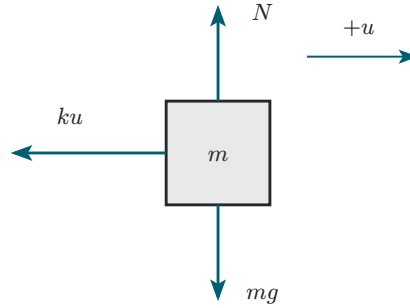
Figure 7: Spring-mass vibrating system.

Consider a spring-mass vibrating system as shown in Figure 7.

Do the following.

- (1) Draw free-body diagram (FBD).
- (2) Write NLM equation.
- (3) Deduce the equation of motion (EOM).
- (4) Deduce the governing equation. Clearly state your assumptions.
- (5) Assume $u(t) = e^{st}$ and substitute into the governing equation.
- (6) Deduce the characteristic equation.
- (7) Define the natural frequency ω_n in terms of m, k .
- (8) Solve the characteristic equation and find $s_{1,2}$ in terms of ω_n .
- (9) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (10) Write the solution in terms of exponential functions.
- (11) Write the solution in terms of trigonometric functions.
- (12) Solve the initial value problem (IVP) and find the response $u(t)$ for:
 - (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to B.1

**Figure 8:** B.1.1. Free-body diagram for the spring-mass system.

I take displacement $u(t)$ as positive to the right and measure it from the equilibrium position. I assume one degree of freedom, a point mass m , a massless linear spring with stiffness k , frictionless horizontal motion, and no damping or external force. Since the spring force opposes the displacement, its horizontal component is

$$\sum F_x = -ku(t)$$

Applying Newton's second law to the mass gives

$$\boxed{\sum F_x = m\ddot{u}(t)} \quad (\text{B.1.2})$$

Equating the spring force and the inertial force produces the equation of motion,

$$\boxed{m\ddot{u}(t) + ku(t) = 0} \quad (\text{B.1.3})$$

Since m is nonzero, division by m gives the normalized governing equation,

$$\boxed{\ddot{u}(t) + \frac{k}{m}u(t) = 0} \quad (\text{B.1.4})$$

Since the equation is linear with constant coefficients, the exponential trial response $u(t) = e^{st}$ applies. Its derivatives are $\dot{u}(t) = se^{st}$ and $\ddot{u}(t) = s^2e^{st}$, so substitution gives

$$\boxed{(ms^2 + k)e^{st} = 0} \quad (\text{B.1.5})$$

Because e^{st} is never zero, the factor multiplying it must vanish. Therefore, the characteristic equation is

$$ms^2 + k = 0 \quad (\text{B.1.6})$$

The undamped natural frequency is defined by

$$\omega_n = \sqrt{\frac{k}{m}} \quad (\text{B.1.7})$$

The characteristic equation then becomes $s^2 + \omega_n^2 = 0$, so its two roots are

$$s_{1,2} = \pm i\omega_n \quad (\text{B.1.8})$$

For positive m and k , these roots have zero real part and nonzero imaginary parts. Thus, the roots are

$$\text{Imaginary} \quad (\text{B.1.9})$$

Using the two distinct roots in the exponential solution gives

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} \quad (\text{B.1.10})$$

Applying Euler's identity converts this response into the equivalent real trigonometric form,

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) \quad (\text{B.1.11})$$

Writing $u_0 = u(0)$ and $\dot{u}_0 = \dot{u}(0)$, evaluation of the trigonometric solution at $t = 0$ gives $A = u_0$. Differentiation followed by evaluation at $t = 0$ gives $\dot{u}_0 = \omega_n B$, so $B = \dot{u}_0 / \omega_n$. Therefore, the three requested initial-condition cases are

$$u(t) = u_0 \cos(\omega_n t) \quad (\text{B.1.12.i})$$

$$u(t) = \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) \quad (\text{B.1.12.ii})$$

$$u(t) = u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) \quad (\text{B.1.12.iii})$$

B.2 Analyze Spring-Mass-Damper Vibrating System

Consider the 1-dof damped system consisting of a spring, k , mass, m , and viscous damper, c , as presented in Figure 9. Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

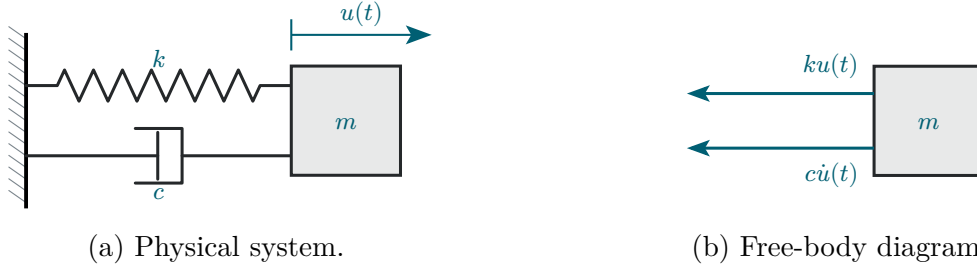


Figure 9: Mass-spring-damper vibrating system.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k .
- (5) Assume $u(t) = e^{st}$ and substitute into EOM.
- (6) Deduce the characteristic equation.
- (7) Find the roots $s_{1,2}$ of the characteristic equation.
- (8) Define critical damping c_{cr} in terms of m and k .
- (9) Define damping ratio ζ .
- (10) Discuss the values of ζ for the three damping cases:
 - (a) Underdamped.
 - (b) Critically damped.
 - (c) Overdamped.
- (11) Express the physical damping c in terms of damping ratio ζ , mass m , and stiffness k .
- (12) Define the natural frequency ω_n in terms of m and k .
- (13) Express in terms of ω_n , ζ the $s_{1,2}$ roots found at (7).
- (14) Assume underdamped system and express the $s_{1,2}$ roots of (13) as complex numbers in terms of ω_n and ζ .
- (15) Define the damped frequency ω_d .
- (16) Substitute (15) into (14) and express the roots s_1 and s_2 in terms of ω_n , ω_d , ζ .
- (17) Assume underdamped conditions and state the inequality that ζ should satisfy.

- (18) Write the general solution of the ODE deduced at (4) in terms of constants C_1 and C_2 multiplying exponential functions of s_1 and s_2 .
- (19) Use (16) to write the solution (18) in terms of exponential functions of ω_n , ω_d , ζ .
- (20) Rewrite the solution (19) in terms of an exponential decay function multiplying the sum of two complex exponentials.
- (21) Rewrite the solution (20) in terms of an exponential decay function multiplying a trigonometric function and some constants C and ϕ to be determined from the initial conditions.
- (22) Sketch the functions discussed at (21).
- (23) Solve the initial value problem (IVP) for $\zeta < 1$ and find the response $u(t)$ for initial displacement u_0 with zero initial velocity.
- (24) Solve the initial value problem (IVP) and find the response $u(t)$ for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to B.2

I take displacement $u(t)$ as positive to the right and measure it from the equilibrium position. I assume one degree of freedom, a point mass m , a massless linear spring with stiffness k , a massless viscous damper with coefficient c , frictionless horizontal motion, and no external force. Since the damping is nonnegative, $c \geq 0$.

The free-body diagram for the damped system is shown below. The spring force opposes displacement, and the damper force opposes velocity.

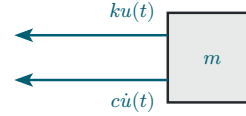


Figure 10: B.2.2. Free-body diagram for the damped system.

Therefore, the spring and damper forces act in the negative x direction, which gives

$$\sum F_x = -c\dot{u}(t) - ku(t) \quad (\text{B.2.2})$$

Applying Newton's law of motion to the mass gives

$$\sum F_x = m\ddot{u}(t) \quad (\text{B.2.3})$$

Equating the force balance to the inertial force and moving all terms to the left gives the equation of motion,

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = 0 \quad (\text{B.2.4})$$

Since the equation is linear with constant coefficients, the nontrivial exponential trial response $u(t) = e^{st}$ applies. Its derivatives are $\dot{u}(t) = se^{st}$ and $\ddot{u}(t) = s^2e^{st}$, so substitution gives

$$(ms^2 + cs + k)e^{st} = 0 \quad (\text{B.2.5})$$

Because e^{st} is never zero, the remaining factor must vanish. Thus, the characteristic equation is

$$ms^2 + cs + k = 0 \quad (\text{B.2.6})$$

Solving this quadratic equation for s gives the two characteristic roots,

$$s_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} \quad (\text{B.2.7})$$

Critical damping occurs when the two roots coincide. Therefore, setting the discriminant equal to zero gives

$$c_{\text{cr}}^2 - 4mk = 0.$$

For nonnegative damping, the critical damping value is

$$\boxed{c_{\text{cr}} = 2\sqrt{mk}} \quad (\text{B.2.8})$$

The damping ratio is the physical damping divided by the critical damping. Hence,

$$\boxed{\zeta = \frac{c}{c_{\text{cr}}} = \frac{c}{2\sqrt{mk}}} \quad (\text{B.2.9})$$

Since $c \geq 0$, the three damping cases are

$$\boxed{0 \leq \zeta < 1} \quad \text{Underdamped} \quad (\text{B.2.10.a})$$

$$\boxed{\zeta = 1} \quad \text{Critically damped} \quad (\text{B.2.10.b})$$

$$\boxed{\zeta > 1} \quad \text{Overdamped} \quad (\text{B.2.10.c})$$

The value $\zeta = 0$ is the undamped limiting case. Using $c = \zeta c_{\text{cr}}$ and the critical damping expression gives the physical damping in terms of ζ , m , and k ,

$$\boxed{c = 2\zeta\sqrt{mk}} \quad (\text{B.2.11})$$

The stiffness-to-mass ratio is the square of the undamped natural frequency. Therefore,

$$\boxed{\omega_n = \sqrt{\frac{k}{m}}} \quad (\text{B.2.12})$$

Using $c/(2m) = \zeta\omega_n$ and $k/m = \omega_n^2$ in the characteristic roots gives

$$\boxed{s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}} \quad (\text{B.2.13})$$

For an underdamped system, $\zeta < 1$, so $\zeta^2 - 1 < 0$. The square root can therefore be written as

$$\sqrt{\zeta^2 - 1} = i\sqrt{1 - \zeta^2}.$$

Substitution into the roots from (B.2.13) gives the complex roots,

$$s_{1,2} = -\zeta\omega_n \pm i\omega_n\sqrt{1-\zeta^2} \quad (\text{B.2.14})$$

The positive imaginary part of the underdamped roots defines the damped frequency,

$$\omega_d = \omega_n\sqrt{1-\zeta^2} \quad (\text{B.2.15})$$

Substituting this definition into (B.2.14) separates the two roots as

$$\begin{aligned} s_1 &= -\zeta\omega_n + i\omega_d, \\ s_2 &= -\zeta\omega_n - i\omega_d. \end{aligned} \quad (\text{B.2.16})$$

The underdamped condition is therefore

$$\zeta < 1 \quad (\text{B.2.17})$$

With physical nonnegative damping, this is $0 \leq \zeta < 1$. The value $\zeta = 0$ gives the undamped limit, while $0 < \zeta < 1$ gives a decaying oscillation.

Using the two roots in (B.2.16), the general solution is the linear superposition of the two exponential solutions,

$$u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t} \quad (\text{B.2.18})$$

Substituting the underdamped roots gives

$$u(t) = C_1 e^{(-\zeta\omega_n + i\omega_d)t} + C_2 e^{(-\zeta\omega_n - i\omega_d)t} \quad (\text{B.2.19})$$

Because $e^{a+b} = e^a e^b$, factor out the common decay term to obtain

$$u(t) = e^{-\zeta\omega_n t} (C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t}) \quad (\text{B.2.20})$$

Euler's identities convert the harmonic part into a real trigonometric form. With $A = C_1 + C_2$ and $B = i(C_1 - C_2)$,

$$C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t} = A \cos(\omega_d t) + B \sin(\omega_d t).$$

Writing this sinusoid in amplitude-phase form gives $A = C \sin \phi$ and $B = C \cos \phi$. Thus,

$$u(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi), \quad C = \sqrt{A^2 + B^2}, \quad \phi = \text{atan2}(A, B) \quad (\text{B.2.21})$$

The sinusoidal carrier is bounded by the two exponential envelopes $u_{\text{env},\pm}(t) = \pm Ce^{-\zeta\omega_n t}$. The carrier keeps the constant damped frequency ω_d , while its amplitude decreases with time. The resulting sketch is shown below using illustrative values $\omega_n = 1$, $\zeta = 0.20$, $C = 1$, and $\phi = 0.5$ radians.

B2. Underdamped response and exponential envelope

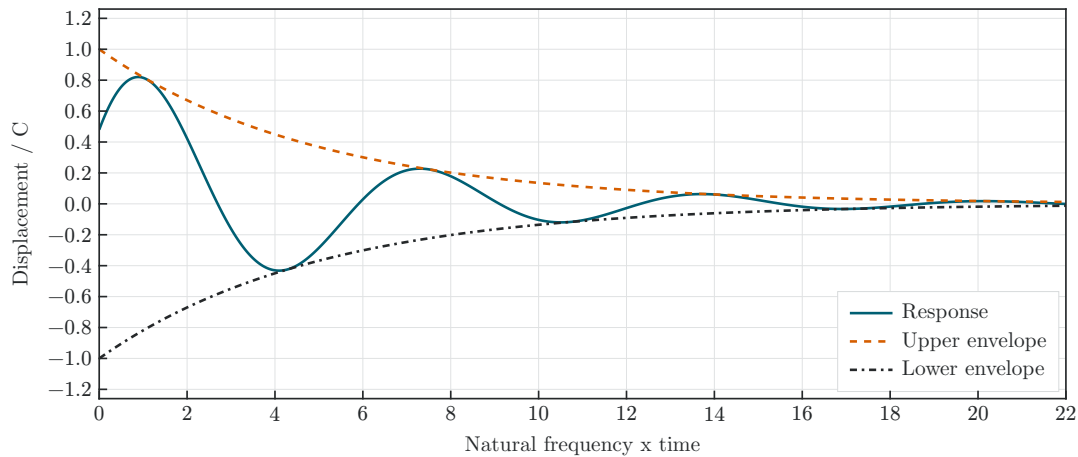


Figure 11: B.2.22. Underdamped oscillation and its exponential decay envelopes.

MATLAB code. Code used to reproduce the B.2.22 response sketch.

```

01 wn = 1; zeta = 0.20; C = 1; phi = 0.5; Nosc = 3;
02 wd = wn*sqrt(1-zeta^2);
03 t = linspace(0,Nosc*2*pi/wd,1000);
04 u = C*exp(-zeta*wn*t).*sin(wd*t+phi);
05 e = C*exp(-zeta*wn*t);
06
07 plot(wn*t,u,'Color',[0 .427 .459]); hold on
08 plot(wn*t,e,'--','Color',[.776 .365 0])
09 plot(wn*t,-e,'-.k')
10 xlabel('Natural frequency x time');
11 ylabel('Displacement / C');
12 legend('Response','Upper envelope','Lower envelope');
13 ylim([-1.2 1.2]); grid on

```

For the initial-displacement case, let $u(0) = u_0$ and $\dot{u}(0) = 0$. Evaluating the cosine-sine form at $t = 0$ gives $A = u_0$. Differentiating that form before setting $t = 0$ gives

$$\dot{u}(0) = -\zeta\omega_n A + \omega_d B = 0.$$

Therefore, $B = \zeta\omega_n u_0 / \omega_d$, and the response is

$$u(t) = u_0 e^{-\zeta\omega_n t} \left[\cos(\omega_d t) + \frac{\zeta\omega_n}{\omega_d} \sin(\omega_d t) \right] \quad (\text{B.2.23})$$

For simultaneous initial displacement and velocity, let $u(0) = u_0$ and $\dot{u}(0) = \dot{u}_0$. The same two initial-condition equations give

$$A = u_0, \quad B = \frac{\dot{u}_0 + \zeta\omega_n u_0}{\omega_d}.$$

Substitution into the trigonometric solution gives the complete underdamped IVP response,

$$u(t) = e^{-\zeta\omega_n t} \left[u_0 \cos(\omega_d t) + \frac{\dot{u}_0 + \zeta\omega_n u_0}{\omega_d} \sin(\omega_d t) \right] \quad (\text{B.2.24})$$

B.3 Calculate Spring-Mass-Damper Response to Initial Displacement

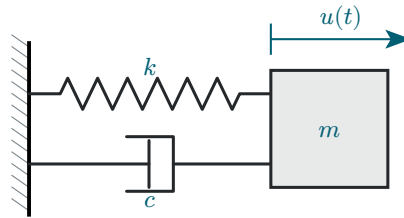


Figure 12: *Spring-mass-damper vibrating system.*

Consider a spring-mass vibrating system with $m = 12 \text{ kg}$, $k = 655 \text{ N/m}$, $c = 19 \text{ N/(m/s)}$ as shown in Figure 12. The vibrating system is given an initial displacement $u_0 = 3.5 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (3) Calculate the critical damping c_{cr} .
- (4) Calculate the damping ratio ζ .
- (5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (6) Calculate the damped period of oscillation τ_d .
- (7) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2 .
- (8) Calculate the constants A , B and then the constants C , ϕ .
- (9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (11) Plot the response of the system for $N_{osc} = 10$ oscillations and use datatips on the plot to find the following:
 - (a) The largest response.
 - (b) The response value calculated at item (9).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10).
- (12) Discuss your results.

Solution to B.3

The given system has mass $m = 12$ kg, stiffness $k = 655$ N/m, damping coefficient $c = 19$ N s/m, initial displacement $u_0 = 3.5$ mm, and initial velocity $\dot{u}_0 = 0$. Using $N_{\text{osc}} = 10$ sets the plotted interval to ten damped oscillations.

Quantity	Value	Unit
m	12.0	kg
k	655	N/m
c	19.0	N s/m
u_0	3.50	mm
$\dot{u}_0$	0	mm/s
N_{osc}	10	–

The natural frequency follows from the stiffness-to-mass ratio. Converting it to hertz requires division by 2π .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{655}{12}} = 7.388 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 1.176 \text{ Hz}.$$

$$\boxed{\omega_n = 7.39 \text{ rad/s}, \quad f_n = 1.176 \text{ Hz}} \quad (\text{B.3.2})$$

Critical damping is the value that makes the characteristic-equation discriminant equal to zero. Therefore,

$$c_{\text{cr}} = 2\sqrt{mk} = 2\sqrt{(12)(655)} = 177.313 \text{ N s/m}.$$

$$\boxed{c_{\text{cr}} = 177.3 \text{ N s/m}} \quad (\text{B.3.3})$$

The damping ratio compares the actual damping with the critical value. Since $c < c_{\text{cr}}$, the system is underdamped.

$$\zeta = \frac{c}{c_{\text{cr}}} = \frac{19}{177.313} = 0.107155.$$

$$\boxed{\zeta = 0.1072 \quad (\text{underdamped})} \quad (\text{B.3.4})$$

For an underdamped system, the oscillatory part of the characteristic roots has frequency $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Converting this frequency to hertz gives

$$\omega_d = 7.388\sqrt{1 - (0.107155)^2} = 7.346 \text{ rad/s}, \quad f_d = \frac{\omega_d}{2\pi} = 1.169 \text{ Hz}.$$

$$\boxed{\omega_d = 7.35 \text{ rad/s}, \quad f_d = 1.169 \text{ Hz}} \quad (\text{B.3.5})$$

One damped oscillation lasts one damped period. Thus,

$$\tau_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{7.346} = 0.8554 \text{ s}.$$

$$\boxed{\tau_d = 0.855 \text{ s}} \quad (\text{B.3.6})$$

The real part of each root is the exponential decay rate $-\zeta\omega_n = -c/(2m) = -0.791667 \text{ s}^{-1}$. Combining this decay rate with the damped frequency gives

$$s_{1,2} = -\zeta\omega_n \pm i\omega_d = -0.791667 \pm i 7.3455 \text{ s}^{-1}.$$

For the exponential form $u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$, the real trigonometric coefficients satisfy $A = C_1 + C_2$ and $B = i(C_1 - C_2)$. The initial conditions determine $A = u_0$ and $B = \zeta\omega_n u_0 / \omega_d$.

$$C_1 = \frac{A - iB}{2} = 1.750 - 0.1886i \text{ mm}, \quad C_2 = \frac{A + iB}{2} = 1.750 + 0.1886i \text{ mm}.$$

$$\boxed{\begin{array}{ll} s_1 = -0.7917 + i 7.3455 \text{ s}^{-1}, & s_2 = -0.7917 - i 7.3455 \text{ s}^{-1}, \\ C_1 = 1.750 - 0.1886i \text{ mm}, & C_2 = 1.750 + 0.1886i \text{ mm} \end{array}} \quad (\text{B.3.7})$$

The zero initial velocity condition fixes the coefficient of the sine term. Substituting the numerical values gives

$$A = u_0 = 3.50 \text{ mm}, \quad B = \frac{\zeta\omega_n u_0}{\omega_d} = 0.3772 \text{ mm}.$$

Writing the harmonic part as a single sine function gives $A = C \sin \phi$ and $B = C \cos \phi$. Therefore,

$$C = \sqrt{A^2 + B^2} = 3.520 \text{ mm}, \quad \phi = \text{atan2}(A, B) = 1.463 \text{ rad}.$$

The displacement response is consequently

$$u(t) = 3.520e^{-0.7917t} \sin(7.3455t + 1.463) \text{ mm.}$$

$$\boxed{\begin{array}{ll} A = 3.50 \text{ mm}, & B = 0.377 \text{ mm}, \\ C = 3.52 \text{ mm}, & \phi = 1.463 \text{ rad} \end{array}} \quad (\text{B.3.8})$$

At one damped period, the sine and cosine terms return to their initial values. The oscillatory factor is therefore one, while the exponential factor has decayed for one period.

$$t_1 = \tau_d = 0.8554 \text{ s}, \quad u_1 = u(t_1) = u_0 e^{-\zeta \omega_n t_1} = 1.778 \text{ mm.}$$

$$\boxed{t_1 = 0.855 \text{ s}, \quad u_1 = 1.778 \text{ mm}} \quad (\text{B.3.9})$$

The displacement magnitude defining the 2% band is

$$u_{0.02} = 0.02u_0 = 0.02(3.5) = 0.0700 \text{ mm.}$$

$$\boxed{u_{0.02} = 0.0700 \text{ mm}} \quad (\text{B.3.10})$$

The largest response occurs at release because the initial displacement is 3.5 mm and the envelope decreases thereafter. The one-period response is the value calculated above. To find the permanent settling time, solve $u(t) = -u_{0.02}$ on the falling side of the last lobe that reaches the lower band. The numerical crossing is $t_s = 4.7867 \text{ s}$. All later extrema have smaller magnitude, so the response remains inside the 2% band after this crossing.

$$\boxed{|u|_{\max} = 3.50 \text{ mm} \quad \text{at } t = 0} \quad (\text{B.3.11.a})$$

$$\boxed{u(t_1) = 1.778 \text{ mm} \quad \text{at } t_1 = 0.855 \text{ s}} \quad (\text{B.3.11.b})$$

$$\boxed{t_s = 4.79 \text{ s}, \quad u(t_s) = -0.0700 \text{ mm}} \quad (\text{B.3.11.c})$$

B3. Displacement release

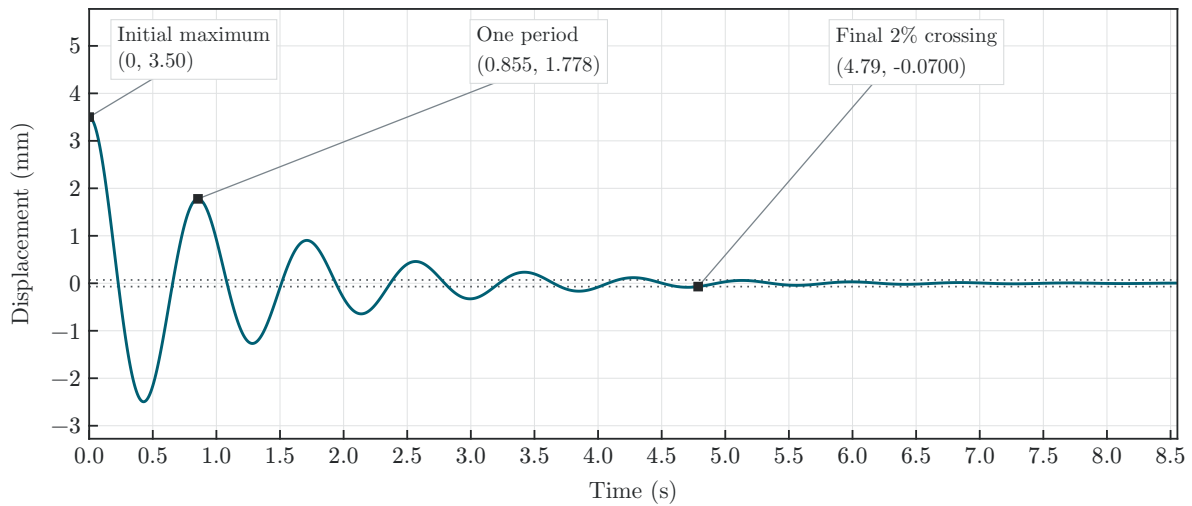


Figure 13: *B.3.11.a-b. Ten-cycle response with the initial maximum, one-period response, and final 2% crossing.*

B3. Final crossing of the 2% band

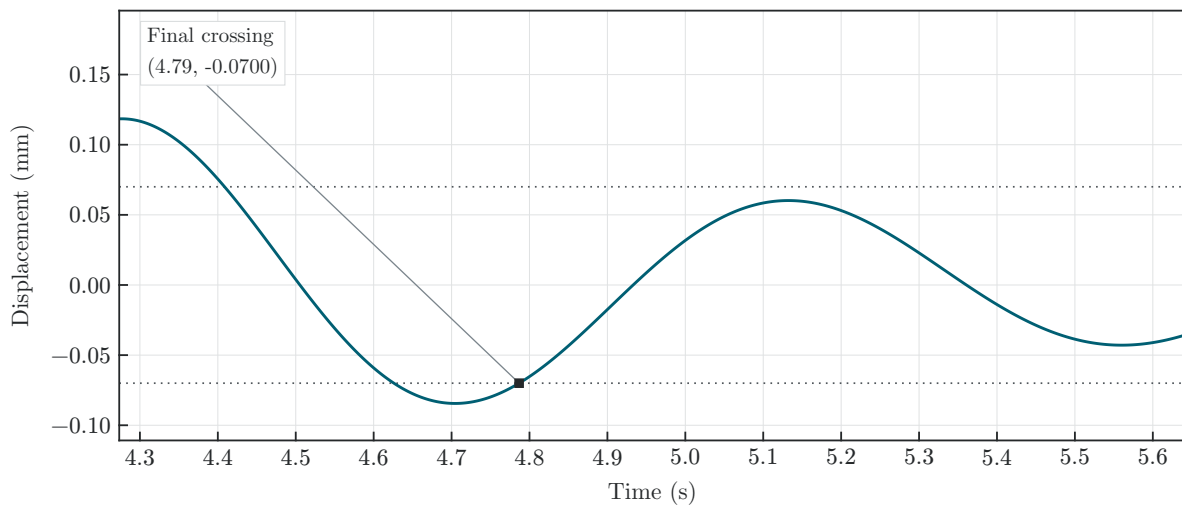


Figure 14: *B.3.11.c. Detail of the final 2% crossing of the settling band.*

MATLAB code. *MATLAB code used to reproduce the B.3.11 response plot.*

```

01 m = 12; k = 655; c = 19; u0 = 3.5; Nosc = 10; ts = 4.786694;
02 wn = sqrt(k/m); zeta = c/(2*sqrt(m*k)); wd = wn*sqrt(1-zeta^2);
03 tau = 2*pi/wd; t = linspace(0,Nosc*tau,2000);
04 u = u0*exp(-zeta*wn*t).*(cos(wd*t)+zeta*wn/wd*sin(wd*t));
05 u02 = .02*u0; u1 = u0*exp(-zeta*wn*tau);
06
07 plot(t,u,'Color',[0 .427 .459]); hold on
08 plot([0 t(end)],[u02 u02],':','Color',[.776 .365 0]);
09 plot([0 t(end)],[-u02 -u02],':','Color',[.776 .365 0]);
10 plot([0 tau ts],[u0 u1 -u02],'ks');
11 xlabel('Time (s)'); ylabel('Displacement (mm)');
12 legend('Response','2% band','Location','northeast'); ylim([-3 5]); grid on

```

The plot shows an underdamped response whose oscillation frequency remains constant while its amplitude decreases exponentially. The initial condition produces the largest displacement at release, and the displacement after one period is smaller by the factor $e^{-\zeta\omega_n\tau_d}$. The response first enters the 2% band before 4.79 s, but it does not remain there until the final crossing at $t_s = 4.79$ s. The exponential envelope gives the conservative bound $t_{\text{envelope}} = 4.95$ s, which is later because the response is not at the envelope maximum at the exact crossing.

(B.3.12)

B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k - m - c spring-mass-damper system as shown in Figure 15.

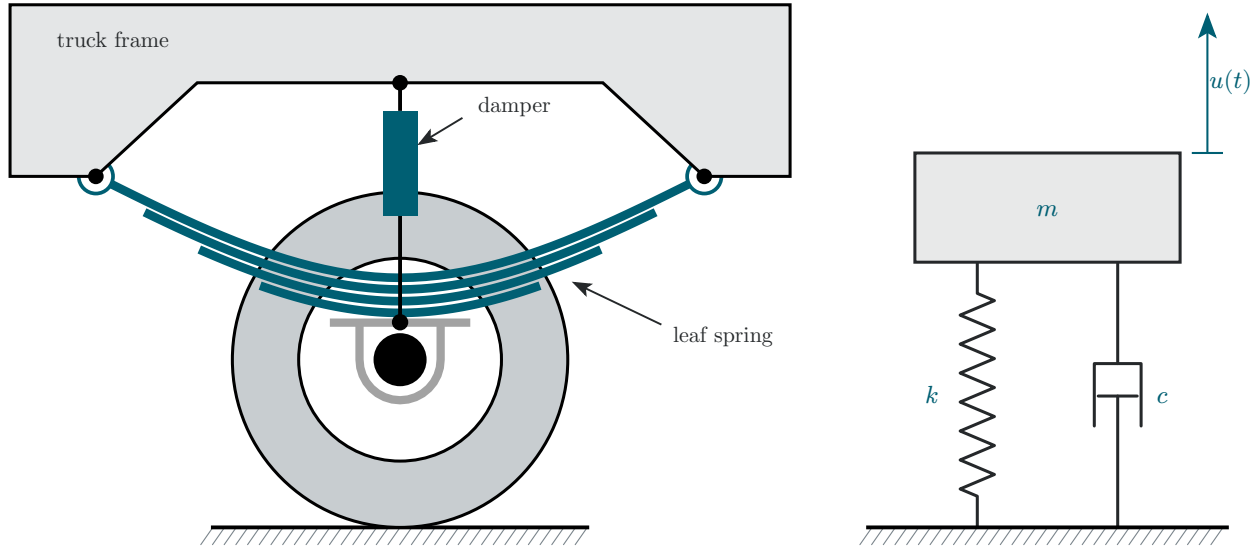


Figure 15: Truck suspension consisting of a leaf spring and an oil damper. The wheel contact is treated as the road datum in the one-degree-of-freedom idealization.

The leaf spring can be approximated by a uniform beam of length $L = 830 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 65 \text{ mm}$, made of steel with $E = 200 \text{ GPa}$. The truck mass apportioned to the wheel can be estimated as $m = 457 \text{ kg}$. Assume initial displacement $u_0 = 125 \text{ mm}$ and zero initial velocity $\dot{u}_0 = 0$.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I of the beam cross section.
- (3) Calculate the flexural stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (6) Calculate the critical damping c_{cr} .
- (7) Assume a 25% damping ratio, i.e., $\zeta = 0.25$, and calculate the physical damping c . Print ζ and c .
- (8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (9) Calculate the damped period of oscillation τ_d .
- (10) Calculate the characteristic-equation roots s_1, s_2 and the constants C_1, C_2 .
- (11) Calculate the constants A, B and then the constants C, ϕ .

-
- (12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (14) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and use datatips on the plot to find the following:
- (a) The largest response.
 - (b) The response value calculated at item (12).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13).
- (15) Discuss your results.
- (16) Graduate work. Assume the damper is worn out such that ζ is reduced to $\zeta = 10\%$.
- (a) Repeat the calculation of items (6) through (12).
 - (b) Compare and discuss the two results.

Solution to B.4

The beam calculation uses SI units, while the response plots report displacement in millimetres. I idealize the leaf spring as a simply supported beam with a concentrated load at its center because this matches the two spring-eye supports and the center displacement shown in the model. Therefore, the corresponding center deflection is

$$\delta = \frac{PL^3}{48EI}, \quad k = \frac{P}{\delta} = \frac{48EI}{L^3}.$$

This support condition is important because a cantilever model would produce a different stiffness.

B.4.1. Input data.

Quantity	Value	Unit
L	0.830	m
b	0.065	m
h	0.003	m
E	200	GPa
m	457	kg
u_0	125	mm
$\dot{u}_0$	0	mm/s
ζ_H	0.25	1
ζ_W	0.10	1
g	9.81	m/s ²

For vertical bending about the horizontal centroidal axis, the section dimension perpendicular to that axis is h , so h is cubed in the second moment of area.

$$I = \frac{bh^3}{12} = \frac{(0.065)(0.003)^3}{12} = 1.4625 \times 10^{-10} \text{ m}^4 = 146.25 \text{ mm}^4.$$

$$\boxed{I = 1.4625 \times 10^{-10} \text{ m}^4 = 146.25 \text{ mm}^4} \quad (\text{B.4.2})$$

Multiplying the section property by the elastic modulus gives the flexural rigidity.

$$EI = (200 \times 10^9)(1.4625 \times 10^{-10}) = 29.25 \text{ N m}^2.$$

$$\boxed{EI = 29.25 \text{ N m}^2} \quad (\text{B.4.3})$$

Substitution into the simply supported center-load relation gives the equivalent spring constant.

$$k = \frac{48EI}{L^3} = \frac{48(29.25)}{(0.830)^3} = 2455.46 \text{ N/m}.$$

$$\boxed{k = 2.455 \times 10^3 \text{ N/m}} \quad (\text{B.4.4})$$

The undamped natural frequency follows from the mass and equivalent stiffness. Its cyclic frequency is found by dividing by 2π .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{2455.46}{457}} = 2.31797 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 0.368917 \text{ Hz}.$$

$$\boxed{\omega_n = 2.32 \text{ rad/s}, \quad f_n = 0.3689 \text{ Hz}} \quad (\text{B.4.5})$$

Critical damping makes the characteristic-equation roots coincide. Therefore,

$$c_{\text{cr}} = 2\sqrt{mk} = 2\sqrt{(457)(2455.46)} = 2118.63 \text{ N s/m}.$$

$$\boxed{c_{\text{cr}} = 2.119 \times 10^3 \text{ N s/m}} \quad (\text{B.4.6})$$

For the original damper, the specified damping ratio is $\zeta_{\text{H}} = 0.25$. The physical damping is the damping ratio multiplied by the critical value.

$$c_{\text{H}} = \zeta_{\text{H}} c_{\text{cr}} = (0.25)(2118.63) = 529.657 \text{ N s/m}.$$

$$\boxed{\zeta_{\text{H}} = 0.250, \quad c_{\text{H}} = 529.7 \text{ N s/m}} \quad (\text{B.4.7})$$

Because $\zeta_{\text{H}} < 1$, the original system is underdamped. Its damped frequency and cyclic frequency are

$$\omega_{d,\text{H}} = \omega_n \sqrt{1 - \zeta_{\text{H}}^2} = 2.31797 \sqrt{1 - 0.25^2} = 2.24437 \text{ rad/s}, \quad f_{d,\text{H}} = \frac{\omega_{d,\text{H}}}{2\pi} = 0.357202 \text{ Hz}.$$

$$\boxed{\omega_{d,\text{H}} = 2.24 \text{ rad/s}, \quad f_{d,\text{H}} = 0.3572 \text{ Hz}} \quad (\text{B.4.8})$$

The time for one damped oscillation is the damped period.

$$\tau_{d,H} = \frac{2\pi}{\omega_{d,H}} = \frac{2\pi}{2.24437} = 2.79954 \text{ s.}$$

$$\boxed{\tau_{d,H} = 2.80 \text{ s}} \quad (\text{B.4.9})$$

Defining the decay rate as $\alpha_H = \zeta_H \omega_n = 0.579493 \text{ s}^{-1}$ gives the characteristic roots

$$s_{1,2} = -\alpha_H \pm i\omega_{d,H} = -0.579493 \pm i 2.24437 \text{ s}^{-1}.$$

For the zero-velocity initial condition, the exponential constants are obtained from the trigonometric coefficients $A = u_0$ and $B = \alpha_H u_0 / \omega_{d,H}$. Since $A = C_1 + C_2$ and $B = i(C_1 - C_2)$,

$$C_1 = \frac{A - iB}{2} = 62.500 - 16.137i \text{ mm}, \quad C_2 = \frac{A + iB}{2} = 62.500 + 16.137i \text{ mm}.$$

$$\boxed{\begin{aligned} s_1 &= -0.5795 + i 2.2444 \text{ s}^{-1}, & s_2 &= -0.5795 - i 2.2444 \text{ s}^{-1}, \\ C_1 &= 62.50 - 16.14i \text{ mm}, & C_2 &= 62.50 + 16.14i \text{ mm} \end{aligned}} \quad (\text{B.4.10})$$

The real trigonometric coefficients are

$$A = u_0 = 125.0 \text{ mm}, \quad B = \frac{\alpha_H u_0}{\omega_{d,H}} = 32.2749 \text{ mm}.$$

Writing the harmonic part as a single sine function gives $A = C \sin \phi$ and $B = C \cos \phi$. Therefore,

$$C = \sqrt{A^2 + B^2} = 129.099 \text{ mm}, \quad \phi = \text{atan2}(A, B) = 1.31812 \text{ rad}.$$

The displacement response for the original damper is consequently

$$u_H(t) = 129.099 e^{-0.579493t} \sin(2.24437t + 1.31812) \text{ mm}.$$

$$\boxed{\begin{aligned} A &= 125.0 \text{ mm}, & B &= 32.27 \text{ mm}, \\ C &= 129.1 \text{ mm}, & \phi &= 1.318 \text{ rad} \end{aligned}} \quad (\text{B.4.11})$$

At one damped period, the sine and cosine terms return to their initial values. The displacement is therefore the initial displacement multiplied by the decay factor for one period.

$$t_{1,H} = \tau_{d,H} = 2.79954 \text{ s}, \quad u_{1,H} = u_H(t_{1,H}) = u_0 e^{-\alpha_H t_{1,H}} = 24.680 \text{ mm}.$$

$$\boxed{t_{1,H} = 2.80 \text{ s}, \quad u_{1,H} = 24.68 \text{ mm}} \quad (\text{B.4.12})$$

The displacement magnitude defining the 2% band is

$$u_{0.02} = 0.02u_0 = 0.02(125) = 2.50 \text{ mm}.$$

$$\boxed{u_{0.02} = 2.50 \text{ mm}} \quad (\text{B.4.13})$$

The largest response is the initial displacement, so $|u|_{\max} = 125 \text{ mm}$ at $t = 0$. The last extremum with magnitude greater than $u_{0.02}$ occurs at approximately $t = 5.599 \text{ s}$, where $u = 4.873 \text{ mm}$. Solving the descending branch for $u(t) = u_{0.02}$ gives the final entry into the 2% band at $t_s = 6.090 \text{ s}$. All later extrema have magnitude below 2.50 mm .

$$\boxed{|u|_{\max} = 125.0 \text{ mm} \quad \text{at } t = 0} \quad (\text{B.4.14.a})$$

$$\boxed{u(t_{1,H}) = 24.68 \text{ mm} \quad \text{at } t_{1,H} = 2.80 \text{ s}} \quad (\text{B.4.14.b})$$

$$\boxed{t_{s,H} = 6.09 \text{ s}, \quad u_H(t_{s,H}) = +2.50 \text{ mm}} \quad (\text{B.4.14.c})$$

B4 healthy. Displacement release

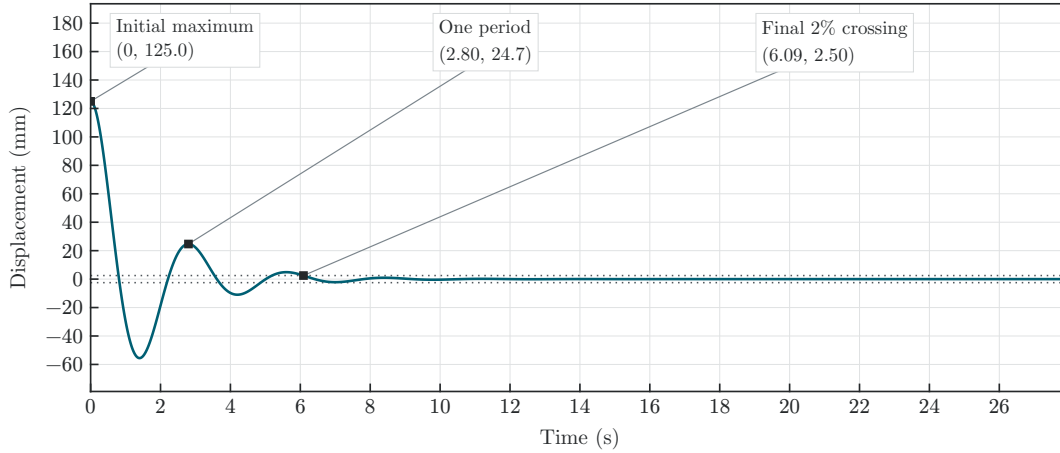


Figure 16: B.4.14.a-b-c. Original damper response for ten oscillations.

B.4.15. Discussion.

The original damper produces a decaying underdamped response. The damped period is 2.80 s, and the displacement after one period is reduced from 125 mm to 24.68 mm. The response remains oscillatory, but its extrema decrease because the envelope is $Ce^{-\alpha_H t}$. The final permanent entry into the 2% band occurs at 6.09 s.

The calculated static displacement under the apportioned weight is

$$\delta_{\text{static}} = \frac{mg}{k} = \frac{(457)(9.81)}{2455.46} = 1.826 \text{ m}.$$

This is larger than the beam span and much larger than the prescribed 125 mm initial displacement. The value indicates that the uniform single-beam model is a formal linear approximation. A physical suspension would require the actual leaf-spring stack, support geometry, preload, and available travel.

For the worn damper, the stiffness and mass do not change. Consequently, I , EI , k , ω_n , f_n , and c_{cr} remain unchanged. Using $\zeta_W = 0.10$ therefore gives the new damping and decay rate

$$c_W = \zeta_W c_{\text{cr}} = 211.863 \text{ N s/m}, \quad \alpha_W = \zeta_W \omega_n = 0.231797 \text{ s}^{-1}.$$

The smaller damping ratio gives

$$\omega_{d,W} = \omega_n \sqrt{1 - \zeta_W^2} = 2.30635 \text{ rad/s}, \quad f_{d,W} = 0.367068 \text{ Hz},$$

$$\tau_{d,W} = \frac{2\pi}{\omega_{d,W}} = 2.72429 \text{ s}.$$

The worn-damper roots and exponential constants are

$$s_{1,2}^W = -0.231797 \pm i 2.30635 \text{ s}^{-1}, \quad C_1^W = 62.500 - 6.281i \text{ mm}, \quad C_2^W = 62.500 + 6.281i \text{ mm}.$$

The corresponding real coefficients are

$$A_W = 125.0 \text{ mm}, \quad B_W = 12.563 \text{ mm}, \quad C_W = 125.630 \text{ mm}, \quad \phi_W = 1.47063 \text{ rad}.$$

Thus, the worn-damper response is

$$u_W(t) = 125.630e^{-0.231797t} \sin(2.30635t + 1.47063) \text{ mm}.$$

The response after one period and the final permanent 2% crossing are

$$t_{1,W} = \tau_{d,W} = 2.72429 \text{ s}, \quad u_{1,W} = u_0 e^{-\alpha_W t_{1,W}} = 66.475 \text{ mm},$$

$$t_{s,W} = 16.559 \text{ s}, \quad u_W(t_{s,W}) = +u_{0.02} = 2.50 \text{ mm}.$$

$c_{cr} = 2118.63 \text{ N s/m},$	$c_W = 211.86 \text{ N s/m},$	
$\omega_{d,W} = 2.306 \text{ rad/s},$	$f_{d,W} = 0.3671 \text{ Hz},$	
$\tau_{d,W} = 2.724 \text{ s},$	$s_{1,2}^W = -0.2318 \pm i 2.306 \text{ s}^{-1},$	
$C_1^W = 62.50 - 6.281i \text{ mm},$	$C_2^W = 62.50 + 6.281i \text{ mm},$	(B.4.16.a)
$A_W = 125.0 \text{ mm},$	$B_W = 12.56 \text{ mm},$	
$C_W = 125.6 \text{ mm},$	$\phi_W = 1.471 \text{ rad},$	
$t_{1,W} = 2.724 \text{ s},$	$u_{1,W} = 66.48 \text{ mm}$	

B.4.16.b. Comparison. The worn damper has a smaller decay rate, so its oscillations persist longer. Its damped frequency is slightly higher and its period is slightly shorter because reducing damping moves ω_d closer to ω_n . The one-period displacement increases from 24.68 mm to 66.48 mm, and the permanent 2% settling time increases from 6.09 s to 16.56 s. Both cases have the same initial maximum because they have the same initial displacement.

B4 worn. Displacement release

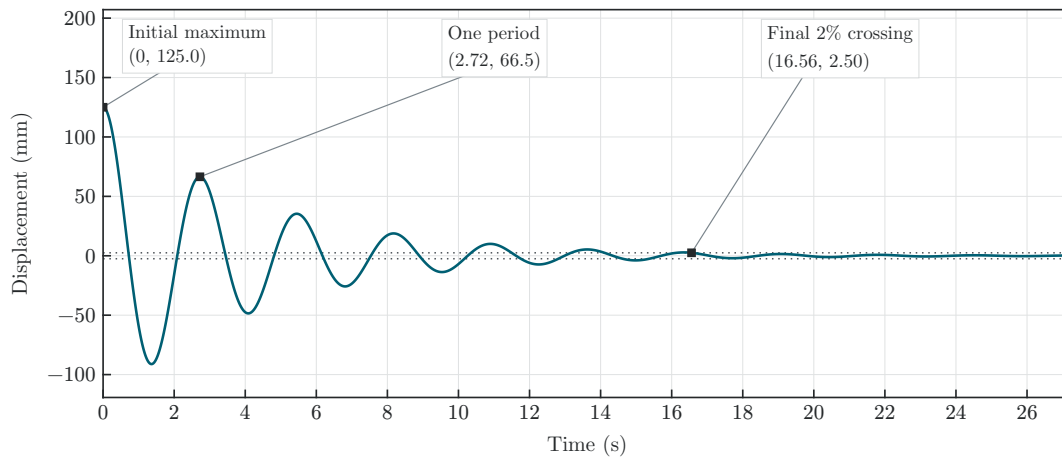


Figure 17: *B.4.16.a. Worn damper response for ten oscillations.*

B4. Worn damping increases the duration of motion

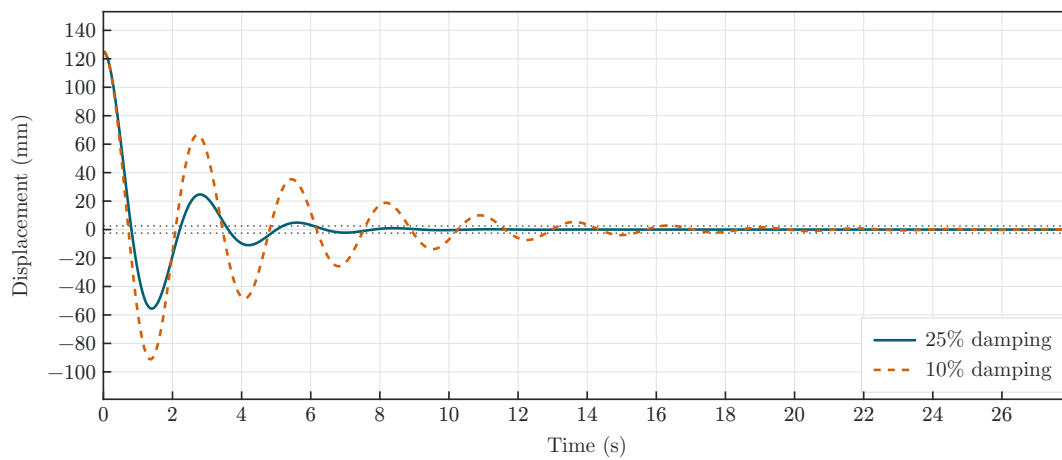


Figure 18: *B.4.16.b. Comparison of the original and worn-damper responses.*

MATLAB code. *MATLAB code used to reproduce the B.4.14 and B.4.16 response plots.*

```

01 L = .830; b = .065; h = .003; E = 200e9; m = 457; u0 = 125;
02 Nosc = 10; zeta = [.25 .10]; ts = [6.090194 16.558987];
03 I = b*h^3/12; k = 48*E*I/L^3; wn = sqrt(k/m);
04
05 for j = 1:2
06     wd = wn*sqrt(1-zeta(j)^2); tau = 2*pi/wd;
07     t = linspace(0,Nosc*tau,2000);
08     u = u0*exp(-zeta(j)*wn*t).*(cos(wd*t) + ...
09         zeta(j)*wn/wd*sin(wd*t));
10     subplot(2,1,j); plot(t,u,'Color',[0 .427 .459]); hold on
11     plot([0 t(end)], [.02*u0 .02*u0],':','Color',[.776 .365 0]);
12     plot([0 t(end)], [-.02*u0 -.02*u0],':','Color',[.776 .365 0]);
13     plot([0 tau ts(j)], [u0 u0*exp(-zeta(j)*wn*tau) .02*u0],'ks');
14     xlabel('Time (s)'); ylabel('Displacement (mm)'); grid on
15 end

```

B.5 Overdamped Spring-Mass-Damper Systems

Consider a spring-mass-damper system with natural frequency $f_n = 6.25$ Hz and various values of the damping ratio ζ . The system is given an initial displacement $u_0 = 13$ mm with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/s frequency ω_n .
- (3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:
 - (a) Write the expressions of the roots s_1, s_2 in terms of ω_n and ζ .
 - (b) Write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2 .
 - (c) Write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Take $\zeta = 2.3$ and calculate the numerical values of s_1, s_2, C_1 , and C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3$ s.
- (4) Assume critically damped conditions $\zeta = 1$ and complete items (a)–(e).
 - (a) Write the expressions of the two partial solutions $u_1(t), u_2(t)$ in terms of ω_n .
 - (b) Write the expression of the complete solution in terms of ω_n and C_1, C_2 .
 - (c) Write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Calculate the numerical values of C_1, C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3$ s.
- (5) Graduate work. Derive the ODE for $\zeta = 1$ and verify that the two solutions $u_1(t)$ and $u_2(t)$ from (4)(a) satisfy the ODE.
- (6) Graduate work. Some race-car manufacturers design for critical damping, whereas others use an overdamped design. Discuss the advantages and disadvantages of the two options.

Solution to B.5

The given initial conditions are $u(0) = u_0 = 13 \text{ mm}$ and $\dot{u}(0) = \dot{u}_0 = 0$. The natural frequency is specified by $f_n = 6.25 \text{ Hz}$.

B.5.1. Input data.

Quantity	Value	Unit
f_n	6.25	Hz
u_0	13.0	mm
$\dot{u}_0$	0	mm/s
t	0–3	s

B.5.2. Natural frequency.

The frequency in hertz is converted to angular frequency by multiplying by 2π .

$$\omega_n = 2\pi f_n = 2\pi(6.25) = 39.2699 \text{ rad/s.}$$

$$\boxed{\omega_n = 39.27 \text{ rad/s}} \quad (\text{B.5.2})$$

B.5.3. Overdamped response with $\zeta = 2.3$.

For free vibration, the characteristic equation is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0.$$

(a) Roots. Since $\zeta > 1$, the discriminant is positive and the two roots are real. Letting s_1 denote the faster root and s_2 the slower root gives

$$s_1 = -\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n, \quad s_2 = -\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n.$$

$$\boxed{s_1 = -\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n, \quad s_2 = -\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n} \quad (\text{B.5.3.a})$$

(b) General solution. Each real root produces an exponential partial solution. Their linear combination is

$$\boxed{u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}} \quad (\text{B.5.3.b})$$

(c) Initial-condition constants. Evaluating the response at $t = 0$ gives $C_1 + C_2 = u_0$. Differentiating first and then evaluating at $t = 0$ gives $s_1 C_1 + s_2 C_2 = \dot{u}_0$. Solving these two equations yields

$$\boxed{C_1 = \frac{\dot{u}_0 - s_2 u_0}{s_1 - s_2}, \quad C_2 = \frac{s_1 u_0 - \dot{u}_0}{s_1 - s_2}} \quad (\text{B.5.3.c})$$

(d) **Numerical values.** For $\zeta = 2.3$,

$$\sqrt{\zeta^2 - 1} = \sqrt{2.3^2 - 1} = 2.07123.$$

Substitution into the root expressions gives

$$\begin{aligned} s_1 &= -(2.3 + 2.07123)(39.2699) = -171.658 \text{ s}^{-1}, \\ s_2 &= -(2.3 - 2.07123)(39.2699) = -8.98372 \text{ s}^{-1}. \end{aligned}$$

Using $u_0 = 13 \text{ mm}$ and $\dot{u}_0 = 0$ in the constant expressions gives

$$C_1 = \frac{0 - (-8.98372)(13)}{-171.658 - (-8.98372)} = -0.71793 \text{ mm},$$

$$C_2 = \frac{(-171.658)(13) - 0}{-171.658 - (-8.98372)} = 13.71793 \text{ mm}.$$

$$\boxed{\begin{aligned} s_1 &= -171.658 \text{ s}^{-1}, & s_2 &= -8.98372 \text{ s}^{-1}, \\ C_1 &= -0.71793 \text{ mm}, & C_2 &= 13.71793 \text{ mm} \end{aligned}} \quad (\text{B.5.3.d})$$

Therefore, the overdamped displacement is

$$u_{\text{OD}}(t) = -0.71793e^{-171.658t} + 13.71793e^{-8.98372t} \text{ mm}.$$

(e) **Overdamped plot.** The response is plotted over the required interval $0 \leq t \leq 3 \text{ s}$.

B5. Critical and overdamped displacement release

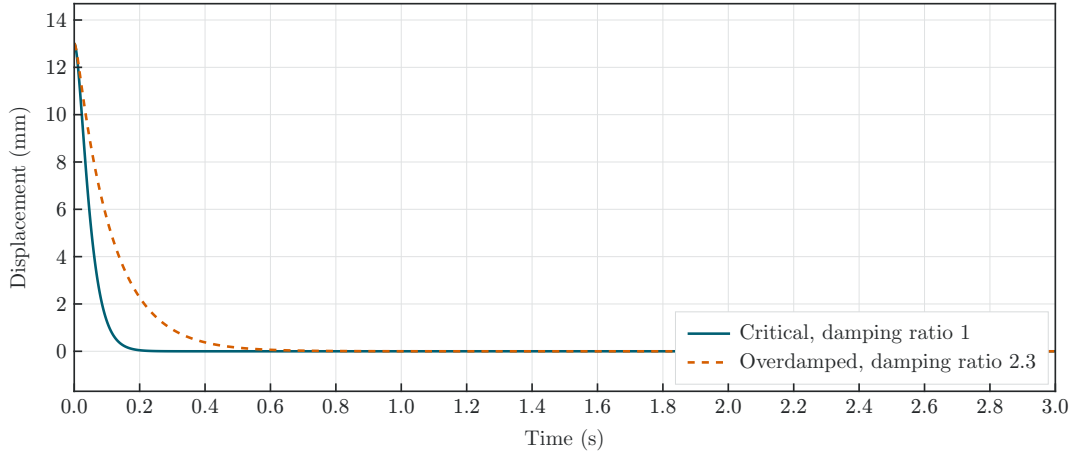


Figure 19: B.5.3.e. Overdamped and critical responses over $0 \leq t \leq 3$ s.

B.5.4. Critically damped response with $\zeta = 1$.

(a) **Partial solutions.** At $\zeta = 1$, the two roots merge into the repeated root $s = -\omega_n$. One solution is the exponential associated with that root. A second independent solution is obtained by multiplying the exponential by t .

$$\boxed{u_1(t) = e^{-\omega_n t}, \quad u_2(t) = t e^{-\omega_n t}} \quad (\text{B.5.4.a})$$

(b) **Complete solution.** The complete response is the linear combination of the two independent partial solutions.

$$\boxed{u(t) = (C_1 + C_2 t) e^{-\omega_n t}} \quad (\text{B.5.4.b})$$

(c) **Initial-condition constants.** At $t = 0$, the displacement condition gives $C_1 = u_0$. Differentiating the complete solution gives

$$\dot{u}(t) = [C_2 - \omega_n(C_1 + C_2 t)] e^{-\omega_n t}.$$

Evaluating at $t = 0$ gives $\dot{u}_0 = C_2 - \omega_n C_1$. Therefore,

$$\boxed{C_1 = u_0, \quad C_2 = \dot{u}_0 + \omega_n u_0} \quad (\text{B.5.4.c})$$

(d) **Numerical values.** The constants are

$$C_1 = 13.000 \text{ mm}, \quad C_2 = 0 + (39.2699)(13) = 510.509 \text{ mm/s}.$$

$$\boxed{C_1 = 13.00 \text{ mm}, \quad C_2 = 510.51 \text{ mm/s}} \quad (\text{B.5.4.d})$$

The critically damped displacement is consequently

$$u_{\text{CD}}(t) = (13 + 510.509t) e^{-39.2699t} \text{ mm.}$$

(e) Critically damped plot. The shorter time view makes the difference between the two nonoscillatory responses visible near the initial displacement.

B5. Critical and overdamped displacement release

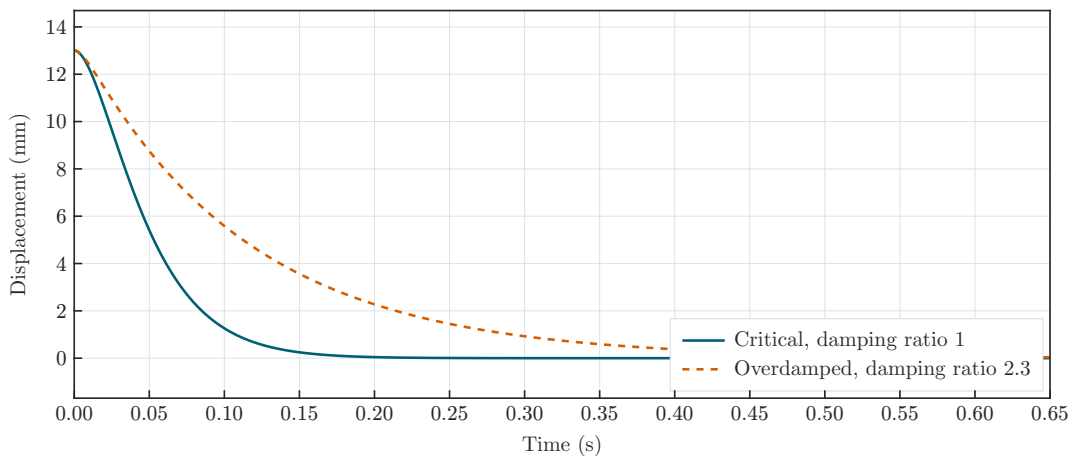


Figure 20: B.5.4.e. Detailed comparison of the critically damped and overdamped responses.

B.5.5. Verification of the critical-damping ODE.

At critical damping, $c = 2m\omega_n$ and $k = m\omega_n^2$. Substitution into $m\ddot{u} + c\dot{u} + ku = 0$, followed by division by m , gives

$$\ddot{u} + 2\omega_n\dot{u} + \omega_n^2u = 0.$$

For $u_1 = e^{-\omega_n t}$, differentiation gives $\dot{u}_1 = -\omega_n u_1$ and $\ddot{u}_1 = \omega_n^2 u_1$. Substitution gives

$$\ddot{u}_1 + 2\omega_n\dot{u}_1 + \omega_n^2u_1 = \omega_n^2u_1 - 2\omega_n^2u_1 + \omega_n^2u_1 = 0.$$

For $u_2 = te^{-\omega_n t}$, the first two derivatives are

$$\dot{u}_2 = (1 - \omega_n t)e^{-\omega_n t}, \quad \ddot{u}_2 = (\omega_n^2 t - 2\omega_n)e^{-\omega_n t}.$$

Substitution into the ODE gives

$$\begin{aligned}\ddot{u}_2 + 2\omega_n\dot{u}_2 + \omega_n^2 u_2 &= (\omega_n^2 t - 2\omega_n)e^{-\omega_n t} + 2\omega_n(1 - \omega_n t)e^{-\omega_n t} + \omega_n^2 t e^{-\omega_n t} \\ &= 0.\end{aligned}$$

Both partial solutions satisfy the critical-damping ODE, so their linear combination is also a solution.

B.5.6. Critical versus overdamped design.

Both designs return to equilibrium without oscillation. Critical damping has the repeated decay rate $-\omega_n$, so it gives the fastest nonoscillatory return for the specified ω_n and initial conditions. Its main disadvantage is that the damping must be tuned close to the critical value. Changes in mass, stiffness, or damper properties move the system away from that condition.

For $\zeta = 2.3$, the overdamped response contains a fast rate of -171.658 s^{-1} and a slow rate of -8.98372 s^{-1} . The fast term reduces the displacement quickly at first, but the slow term controls the long tail and makes the final settling slower. Overdamping can therefore provide more tolerance against crossing into oscillatory behavior, but it sacrifices response speed and can increase damper forces during rapid motion. A race-car suspension must balance these effects with tire contact, road inputs, suspension travel, and damper-force limits.

MATLAB code. *MATLAB script for the B.5 overdamped and critically damped response plots.*

```
01 fn = 6.25; u0 = 13; v0 = 0; z = 2.3;
02 t = linspace(0,3,2000); wn = 2*pi*fn;
03
04 q = sqrt(z^2-1); s1 = -(z+q)*wn; s2 = -(z-q)*wn;
05 C1 = (v0-s2*u0)/(s1-s2); C2 = (s1*u0-v0)/(s1-s2);
06 uOD = C1*exp(s1*t)+C2*exp(s2*t);
07 uCD = (u0+(v0+wn*u0)*t).*exp(-wn*t);
08
09 plot(t,uCD,'Color',[0 .427 .459],'LineWidth',1.5); hold on
10 plot(t,uOD,'--','Color',[.776 .365 0],'LineWidth',1.5)
11 xlabel('Time (s)'); ylabel('Displacement (mm)'); grid on
12 legend('Critical, zeta = 1','Overdamped, zeta = 2.3')
```

B.6 Analyze Spring-Mass-Damper Forced Vibration

Consider a mass-spring-damper vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 21.

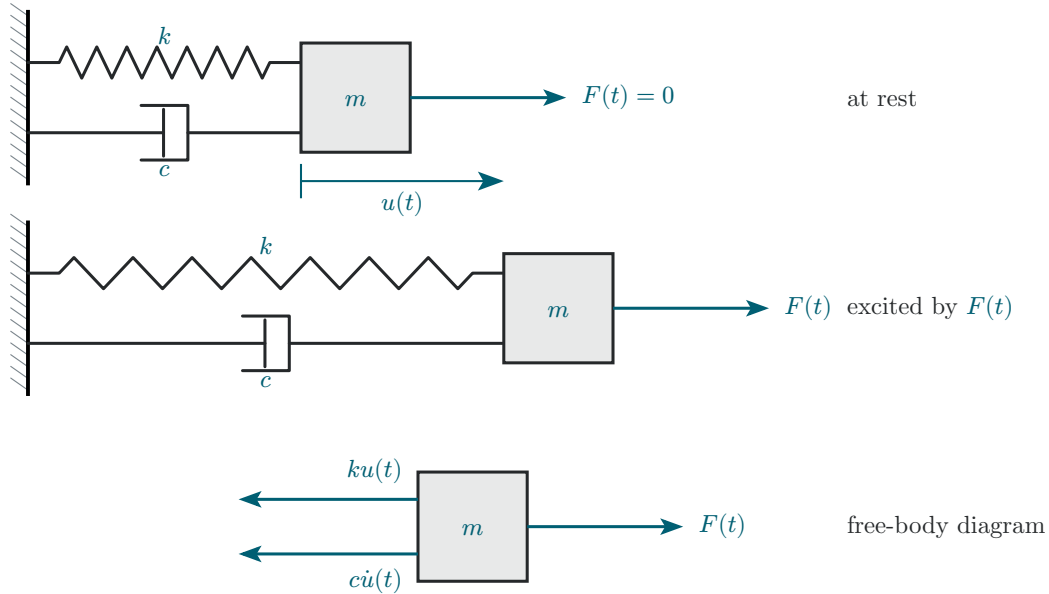


Figure 21: Spring-mass-damper system excited by force $F(t)$.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k , $F(t)$.
- (5) Assume harmonic excitation $F(t) = \hat{F} e^{i\omega t}$, substitute in (4), and divide by m .
- (6) Define the normalized excitation amplitude $\hat{f} = \hat{F}/m$ and recall the definitions of ω_n and ζ to obtain a nonhomogeneous linear ODE in ω_n , ζ , $\hat{f}$.
- (7) Write the general solution $u(t)$ as a sum of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (8) Write the general form of the complementary solution $u_c(t)$.
- (9) Find the particular solution $u_p(t)$.
- (10) Use (8), (9) to write the complete solution $u(t)$; discuss the meaning of transient response and steady-state response.
- (11) Find an expression for the steady-state response amplitude $\hat{u}_{ss}$ in terms of ω_n , ζ , $\hat{f}$, and ω .
- (12) Develop the expression for the frequency response function $\text{FRF}(\omega)$ in terms of ω_n , ζ , ω .
- (13) Discuss the variation with frequency of FRF magnitude and phase.

- (14) Give the definition of the magnification factor $M(\omega)$.
- (15) Give the expression for the magnification factor $M(\omega)$ in terms of ω_n and ζ .
- (16) Give the definition of the resonance frequency ω_r .
- (17) Give the expression for the resonance frequency ω_r in terms of ω_n and ζ .
- (18) Give the expression of the peak magnification M_r .
- (19) Give the expression for the FRF phase $\phi(p)$, where $p = \omega/\omega_n$.
- (20) Give the definition of phase resonance and deduce the FRF expression at phase resonance.
- (21) Calculate the magnification factor at phase resonance.
- (22) Calculate the magnitude of the physical response at phase resonance.
- (23) Discuss the difference between true resonance and phase resonance.
- (24) Write and compare the expressions for the resonance frequency ω_r and damped frequency ω_d .
- (25) Establish a formula to calculate the RPM corresponding to a rad/sec frequency ω .

Solution to B.6

B.6.1. Assumptions. I assume one translational degree of freedom $u(t)$, a point mass m , a linear spring with stiffness k , and a viscous damper with coefficient c . I also assume a fixed support, small motion, and constant parameters m , c , and k . The applied force $F(t)$ acts in the positive x direction, while the spring and damper forces oppose the motion.

B.6.2. Free-body diagram and force sum. I take positive x to the right. Since the applied force $F(t)$ acts to the right while the spring and damper reactions act to the left,

$$\sum F_x = F(t) - c\dot{u}(t) - ku(t) \quad (\text{B.6.2})$$

The corresponding isolated mass is shown below.

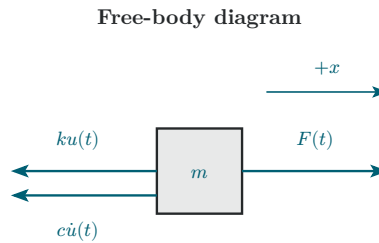


Figure 22: B.6.2. Free-body diagram for the forced spring-mass-damper system.

B.6.3. Newton's law of motion. Newton's law relates the net force to the mass acceleration.

$$\sum F_x = m\ddot{u}(t) \quad (\text{B.6.3})$$

B.6.4. Equation of motion. Equating the force sum to the inertial term gives

$$F(t) - c\dot{u}(t) - ku(t) = m\ddot{u}(t).$$

Moving the spring and damper terms to the left gives the governing ODE.

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = F(t) \quad (\text{B.6.4})$$

B.6.5. Harmonic excitation and division by mass. Representing the applied force by a complex harmonic function gives

$$F(t) = \hat{F}e^{i\omega t}.$$

Substitution into the equation of motion and division by m gives

$$\ddot{u} + \frac{c}{m}\dot{u} + \frac{k}{m}u = \frac{\hat{F}}{m}e^{i\omega t} \quad (\text{B.6.5})$$

The physical force and displacement are obtained by taking the real parts of the complex expressions.

B.6.6. Normalized nonhomogeneous ODE. The normalized force amplitude and natural frequency are defined by

$$\hat{f} = \frac{\hat{F}}{m}, \quad \omega_n = \sqrt{\frac{k}{m}}.$$

The damping ratio is

$$\zeta = \frac{c}{2\sqrt{mk}} = \frac{c}{2m\omega_n}, \quad \frac{c}{m} = 2\zeta\omega_n, \quad \frac{k}{m} = \omega_n^2.$$

Substitution into the divided equation produces the normalized forced-vibration ODE.

$$\ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2u = \hat{f}e^{i\omega t} \quad (\text{B.6.6})$$

B.6.7. General solution. The right-hand side is nonzero, so the solution is the sum of a complementary solution and a particular solution.

$$u(t) = u_c(t) + u_p(t) \quad (\text{B.6.7})$$

The complementary part satisfies the homogeneous equation, and the particular part accounts for the harmonic force.

B.6.8. Complementary solution. For the underdamped case $0 \leq \zeta < 1$, define the damped frequency as

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}.$$

The homogeneous equation then has the real solution

$$u_c(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (\text{B.6.8})$$

where C and ϕ are determined by the initial conditions after the particular solution is included. Equivalently, the same response can be written as

$$u_c(t) = e^{-\zeta\omega_n t} [A \cos(\omega_d t) + B \sin(\omega_d t)].$$

B.6.9. Particular solution. Because the excitation is proportional to $e^{i\omega t}$, seek a particular solution in the same form,

$$u_p(t) = \hat{u}_p e^{i\omega t}.$$

Differentiation gives $\dot{u}_p = i\omega \hat{u}_p e^{i\omega t}$ and $\ddot{u}_p = -\omega^2 \hat{u}_p e^{i\omega t}$. Substitution into the normalized ODE gives

$$(-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2) \hat{u}_p = \hat{f}.$$

Solving for the complex displacement amplitude gives

$$\hat{u}_p = \frac{\hat{f}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega}, \quad u_p(t) = \frac{\hat{f}e^{i\omega t}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \quad (\text{B.6.9})$$

B.6.10. Complete response and its meaning. Combining the complementary and particular parts gives

$$u(t) = C e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) + \frac{\hat{f}e^{i\omega t}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \quad (\text{B.6.10})$$

The first term is the transient response. It decays because of the factor $e^{-\zeta\omega_n t}$. The second term is the steady-state response. It remains after the transient has decayed and oscillates at the forcing frequency ω . Initial conditions must be applied to the complete response, since both terms contribute at the start of the motion.

B.6.11. Steady-state response amplitude. The complex amplitude is the coefficient multiplying $e^{i\omega t}$ in the particular response.

$$\hat{u}_{ss}(\omega) = \frac{\hat{f}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \quad (\text{B.6.11})$$

Using $\hat{f} = \hat{F}/m$ gives the equivalent physical-parameter form

$$\hat{u}_{ss}(\omega) = \frac{\hat{F}}{k - m\omega^2 + i c \omega}.$$

B.6.12. Frequency response function. The quasi-static displacement is obtained by taking $\omega \rightarrow 0$.

$$u_{\text{qst}} = \frac{\hat{F}}{k} = \frac{\hat{f}}{\omega_n^2}.$$

Normalize the steady-state amplitude by this displacement to obtain

$$\boxed{\text{FRF}(\omega) = \frac{\hat{u}_{\text{ss}}}{u_{\text{qst}}} = \frac{\omega_n^2}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega}} \quad (\text{B.6.12})$$

With $p = \omega/\omega_n$, this becomes

$$\text{FRF}(p) = \frac{1}{1 - p^2 + 2i\zeta p}.$$

B.6.13. FRF variation with frequency. At zero excitation frequency, $\text{FRF}(0) = 1$, so the response equals the quasi-static displacement and has zero phase lag. As p increases, the magnitude can rise near the natural-frequency range and then decreases toward zero. The phase is negative because the displacement lags the applied force. It changes from 0 at low frequency toward $-\pi$ as p becomes large. Increasing ζ lowers and broadens the magnitude peak. When $\zeta \geq 1/\sqrt{2}$, the magnitude has no nonzero interior peak.

B.6.14. Magnification-factor definition. The magnification factor is the magnitude of the normalized steady-state response.

$$\boxed{M(\omega) = |\text{FRF}(\omega)| = \frac{|\hat{u}_{\text{ss}}|}{u_{\text{qst}}}} \quad (\text{B.6.14})$$

B.6.15. Magnification factor in terms of ω_n and ζ . Taking the magnitude of the normalized FRF gives

$$\boxed{M(\omega) = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}} = \frac{1}{\sqrt{(1 - p^2)^2 + (2\zeta p)^2}}} \quad (\text{B.6.15})$$

B.6.16. Resonance-frequency definition. The true resonance frequency is the frequency at which the magnification factor is maximum.

$$\boxed{\omega_r = \arg \max_{\omega \geq 0} M(\omega)} \quad (\text{B.6.16})$$

B.6.17. Resonance frequency expression. It is convenient to minimize the reciprocal squared magnitude,

$$M^{-2}(p) = Q(p) = (1 - p^2)^2 + 4\zeta^2 p^2.$$

Differentiation gives

$$\frac{dQ}{dp} = 4p (p^2 + 2\zeta^2 - 1).$$

The nonzero stationary point exists when $0 \leq \zeta < 1/\sqrt{2}$, which gives

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2}, \quad 0 \leq \zeta < \frac{1}{\sqrt{2}} \quad (\text{B.6.17})$$

For $\zeta \geq 1/\sqrt{2}$, the square root is not real and the maximum occurs at the zero-frequency limit rather than at a nonzero resonance frequency.

B.6.18. Peak magnification. Since $p_r^2 = 1 - 2\zeta^2$ at true resonance, substitution into the magnification expression gives

$$M_r = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}, \quad 0 < \zeta < \frac{1}{\sqrt{2}} \quad (\text{B.6.18})$$

For zero damping, the ideal model has an unbounded response at $\omega = \omega_n$.

B.6.19. FRF phase. The phase is the argument of the complex FRF. Using the quadrant-aware inverse tangent gives

$$\phi(p) = \arg[\text{FRF}(p)] = -\text{atan2}(2\zeta p, 1 - p^2) \quad (\text{B.6.19})$$

This expression gives $\phi(0) = 0$ and approaches $-\pi$ as $p \rightarrow \infty$.

B.6.20. Phase resonance. Phase resonance is the condition that the displacement lags the applied force by 90° , or $\phi = -\pi/2$. The real part of the FRF denominator must then vanish,

$$1 - p^2 = 0.$$

For positive frequency, this gives $p = 1$ and $\omega = \omega_n$. At this frequency,

$$\text{FRF}(\omega_n) = \frac{1}{2i\zeta} = -\frac{i}{2\zeta} \quad (\text{B.6.20})$$

B.6.21. Magnification at phase resonance. Taking the magnitude of the phase-resonance FRF gives

$$M_{90} = |\text{FRF}(\omega_n)| = \frac{1}{2\zeta} \quad (\text{B.6.21})$$

B.6.22. Physical response at phase resonance. Multiplying the normalized FRF by the

quasi-static displacement gives the complex response amplitude,

$$\hat{u}_{90} = u_{\text{qst}} \text{FRF}(\omega_n) = -i \frac{\hat{F}}{2\zeta k}.$$

Therefore, the physical response magnitude is

$$|\hat{u}_{90}| = \frac{\hat{F}}{2\zeta k} = \frac{\hat{F}}{c\omega_n} \quad (\text{B.6.22})$$

B.6.23. True resonance versus phase resonance. True resonance identifies the frequency at which the response magnitude is maximum. For light damping, it occurs at $\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$, which is below ω_n . Phase resonance identifies the frequency at which the response lags by 90° , and it occurs at $\omega = \omega_n$ for every positive damping ratio. These frequencies are equal only in the zero-damping limit. For $\zeta \geq 1/\sqrt{2}$, there is no nonzero true-resonance peak, but the phase-resonance condition still occurs at ω_n .

B.6.24. Resonance and damped frequencies. For an underdamped system,

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}.$$

When $0 < \zeta < 1/\sqrt{2}$, the three frequencies satisfy

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} < \omega_d = \omega_n \sqrt{1 - \zeta^2} < \omega_n \quad (\text{B.6.24})$$

Thus, the true resonance frequency is below the damped free-vibration frequency, which is below the undamped natural frequency.

B.6.25. Conversion from rad/s to RPM. One complete excitation cycle per revolution corresponds to an angular speed of 2π radians per revolution. Therefore, a frequency ω in rad/s corresponds to

$$N_{\text{RPM}} = \frac{60\omega}{2\pi} = \frac{30\omega}{\pi} \text{ RPM} \quad (\text{B.6.25})$$

B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation

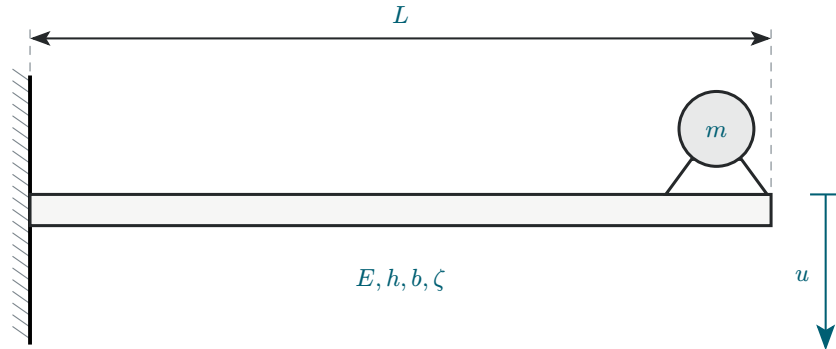


Figure 23: Damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor.

Consider a cantilever beam of length $L = 568 \text{ mm}$, elastic modulus $E = 200 \text{ GPa}$, cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$, and damping ratio $\zeta = 14\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m = 0.52 \text{ kg}$ operating at $1,500 \text{ RPM}$ is attached at the beam tip as shown in Figure 23. The electric motor is slightly unbalanced, producing a slight vibratory excitation $F(t) = \hat{F}e^{i\omega t}$ with $\hat{F} = 12 \text{ mN}$. The excitation frequency ω depends on the electric motor speed.

Do the following.

- (1) Display input data.
- (2) Calculate:
 - (a) Cross-section second moment of area I , flexural stiffness EI , and spring constant k .
 - (b) Natural frequency ω_n rad/s, critical damping c_{cr} , physical damping c .
 - (c) Quasi-static deflection u_{qst} corresponding to $\omega_n \rightarrow 0$.
- (3) Calculate the values of:
 - (a) Resonant frequency ω_r and damped frequency ω_d .
 - (b) Peak magnification M_r .
 - (c) Magnification at phase resonance M_{90} .
 - (d) Physical response $\hat{u}_{90}$ at phase resonance and its magnitude $|\hat{u}_{90}|$.
 - (e) Calculate the Hz frequencies f_r , f_n and the corresponding RPM values.
- (4) Assume excitation at phase resonance:
 - (a) Calculate the excitation frequency ω in rad/s and f in Hz.
 - (b) Calculate period of oscillation τ .
 - (c) Plot the time response for $N_{osc} = 10$ oscillations. Hint: plot the real part of the complex displacement $u(t)$.
- (5) Graduate work. Complete items (a)–(d).

- (a) Calculate and display the operational frequency f_0 in Hz and the corresponding ω_0 in rad/sec. Plot the magnitude and phase of the frequency response function FRF vs. excitation f in Hz from quasi-static to f_0 with $N_f = 5000$ points.
- (b) Calculate and display the FRF magnitude at phase-resonance RPM, at double the phase resonance RPM, and at operational RPM. Display the corresponding Hz frequencies. Put the corresponding data tips on the magnitude plot.
- (c) Calculate the vibration magnitude $|u_0|$ when the electric motor runs at its operational RPM.
- (d) Comment on what vibration would be observed as the motor is switched on and accelerates to reach its operational RPM.

Solution to B.7

B.7.1. Input data. I model the cantilever by its equivalent tip stiffness and use the attached motor mass as the lumped vibrating mass. I omit beam mass because neither beam density nor beam mass is supplied. Since h is the bending depth for vertical bending, the second moment of area is $I = bh^3/12$.

Quantity	Value	Unit
L	0.568	m
E	200	GPa
h	3.00	mm
b	37.0	mm
m	0.520	kg
ζ	0.140	1
$\hat{F}$	12.0	mN
N_0	1500	rpm
N_{osc}	10	oscillations
N_f	5000	points

B.7.2. Beam and damping parameters.

(a) Section and beam stiffness. Because the beam bends about the weak axis, the rectangular-section second moment of area is

$$I = \frac{bh^3}{12} = \frac{(0.037)(0.003)^3}{12} = 8.325 \times 10^{-11} \text{ m}^4 = 83.25 \text{ mm}^4.$$

The tip stiffness of a fixed-free beam is $k = 3EI/L^3$. Therefore,

$$EI = (200 \times 10^9)(8.325 \times 10^{-11}) = 16.65 \text{ N m}^2, \quad k = \frac{3(16.65)}{(0.568)^3} = 272.58 \text{ N/m}.$$

$$\boxed{I = 8.325 \times 10^{-11} \text{ m}^4, \quad EI = 16.65 \text{ N m}^2, \quad k = 272.58 \text{ N/m}} \quad (\text{B.7.2.a})$$

(b) Natural frequency and damping. The natural frequency follows from the lumped mass and beam stiffness.

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{272.58}{0.520}} = 22.895 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 3.6439 \text{ Hz}.$$

Critical damping is $c_{\text{cr}} = 2m\omega_n$. Since the damping ratio is $\zeta = c/c_{\text{cr}}$, the physical damping coefficient is

$$c_{\text{cr}} = 2(0.520)(22.895) = 23.811 \text{ N s/m}, \quad c = \zeta c_{\text{cr}} = (0.140)(23.811) = 3.3335 \text{ N s/m}.$$

$$\begin{aligned} \omega_n &= 22.895 \text{ rad/s}, & f_n &= 3.6439 \text{ Hz}, \\ c_{\text{cr}} &= 23.811 \text{ N s/m}, & c &= 3.3335 \text{ N s/m} \end{aligned} \quad (\text{B.7.2.b})$$

(c) Quasi-static deflection. Because quasi-static loading makes the excitation frequency negligible, $\omega \rightarrow 0$ while ω_n remains fixed. The force-normalized frequency response then approaches one, so

$$u_{\text{qst}} = \frac{\hat{F}}{k} = \frac{0.012}{272.58} = 4.4024 \times 10^{-5} \text{ m.}$$

$$u_{\text{qst}} = 44.024 \text{ } \mu\text{m} \quad (\text{B.7.2.c})$$

B.7.3. Resonance quantities.

(a) True resonance and damped frequency. Since $\zeta < 1/\sqrt{2}$, the magnification has a nonzero maximum. Differentiating the magnification with respect to the frequency ratio gives

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} = 22.442 \text{ rad/s.}$$

The damped natural frequency comes from the complementary solution.

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 22.670 \text{ rad/s.}$$

$$\omega_r = 22.442 \text{ rad/s}, \quad \omega_d = 22.670 \text{ rad/s} \quad (\text{B.7.3.a})$$

(b) Peak magnification. Substituting ω_r into the magnification expression gives

$$M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} = \frac{1}{2(0.140)\sqrt{1 - (0.140)^2}} = 3.60695.$$

$$M_r = 3.60695 \quad (\text{B.7.3.b})$$

(c) Magnification at phase resonance. Phase resonance occurs when the real part of the FRF denominator is zero, which gives $\omega = \omega_n$. Thus,

$$M_{90} = \frac{1}{2\zeta} = \frac{1}{2(0.140)} = 3.57143.$$

$$\boxed{M_{90} = 3.57143} \quad (\text{B.7.3.c})$$

(d) **Physical response at phase resonance.** At $\omega = \omega_n$, the normalized FRF is

$$\text{FRF}(\omega_n) = \frac{1}{2i\zeta} = -\frac{i}{2\zeta}.$$

Multiplying by the quasi-static displacement gives

$$\hat{u}_{90} = u_{\text{qst}} \text{FRF}(\omega_n) = -i \frac{u_{\text{qst}}}{2\zeta} = -i 157.229 \mu\text{m}.$$

Therefore,

$$\boxed{\hat{u}_{90} = -i 157.229 \mu\text{m}, \quad |\hat{u}_{90}| = 157.229 \mu\text{m}} \quad (\text{B.7.3.d})$$

(e) **Frequency and RPM conversions.** Since $f = \omega/(2\pi)$, division by 2π converts each angular frequency to hertz. Multiplication by 60 then converts hertz to revolutions per minute.

$$\begin{aligned} f_r &= \frac{\omega_r}{2\pi} = 3.57174 \text{ Hz}, & N_r &= 60f_r = 214.305 \text{ rpm}, \\ f_n &= \frac{\omega_n}{2\pi} = 3.64388 \text{ Hz}, & N_{90} &= 60f_n = 218.633 \text{ rpm}. \end{aligned}$$

$$\boxed{\begin{aligned} f_r &= 3.57174 \text{ Hz}, & N_r &= 214.305 \text{ rpm}, \\ f_n &= 3.64388 \text{ Hz}, & N_{90} &= 218.633 \text{ rpm} \end{aligned}} \quad (\text{B.7.3.e})$$

B.7.4. Time response at phase resonance.

(a) **Excitation frequency.** Phase resonance uses $\omega = \omega_n$, so

$$\boxed{\omega = 22.895 \text{ rad/s}, \quad f = \frac{\omega}{2\pi} = 3.64388 \text{ Hz}} \quad (\text{B.7.4.a})$$

(b) **Period.** The period of the phase-resonant oscillation is

$$\tau = \frac{2\pi}{\omega} = \frac{2\pi}{22.895} = 0.274433 \text{ s}.$$

$$\boxed{\tau = 0.274433 \text{ s}, \quad N_{\text{osc}}\tau = 2.74433 \text{ s}} \quad (\text{B.7.4.b})$$

(c) **Real displacement.** The physical force is taken as $F(t) = \hat{F} \cos(\omega_n t)$. Since $\hat{u}_{90} = -iU$, where

$$U = \frac{u_{\text{qst}}}{2\zeta} = 157.229 \text{ } \mu\text{m},$$

the real part of the complex steady-state response is

$$u_{\text{ss}}(t) = \text{Re}\{\hat{u}_{90}e^{i\omega_n t}\} = U \sin(\omega_n t).$$

The assignment specifies the number of cycles but not initial conditions for this plot, so the steady-state response is the direct result. For comparison, the figure also shows the response when the force is switched on at $t = 0$ and the mass starts from rest. Applying $u(0) = 0$ and $\dot{u}(0) = 0$ to the complete response gives

$$u_{\text{rest}}(t) = U \sin(\omega_n t) - \frac{U\omega_n}{\omega_d} e^{-\zeta\omega_n t} \sin(\omega_d t).$$

B7. Phase-resonant response

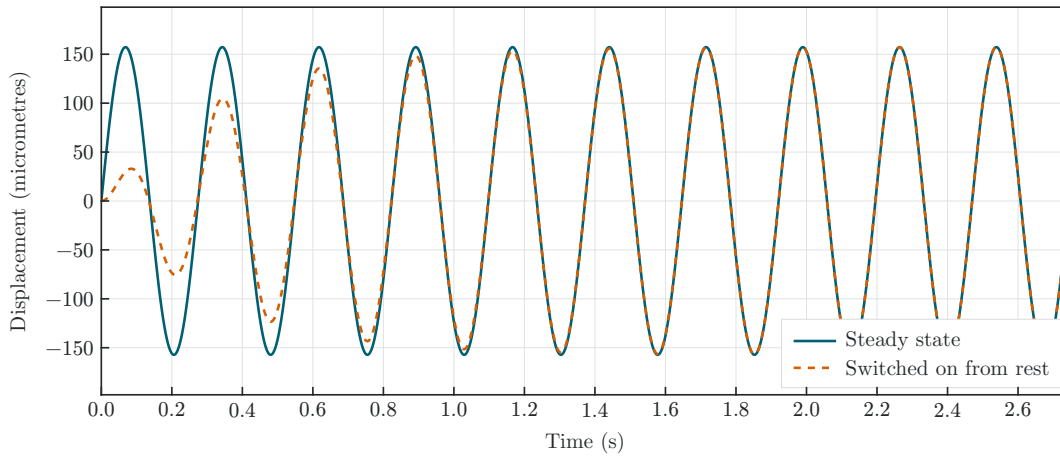


Figure 24: B.7.4.c. Phase-resonant real displacement over $N_{\text{osc}} = 10$ cycles. The orange curve includes the transient required when the force is switched on from rest.

B.7.5. Graduate work.

(a) **Operational frequency and FRF sweep.** The operating speed is $N_0 = 1500$ rpm, so

$$f_0 = \frac{N_0}{60} = 25.0 \text{ Hz}, \quad \omega_0 = 2\pi f_0 = 157.080 \text{ rad/s}.$$

For the sweep, define $p = \omega/\omega_n = 2\pi f/\omega_n$. The magnitude and phase are

$$M(f) = \frac{1}{\sqrt{(1-p^2)^2 + (2\zeta p)^2}}, \quad \phi(f) = -\text{atan2}(2\zeta p, 1-p^2).$$

Using $N_f = 5000$ uniformly spaced frequencies from $f = 0$ to $f = f_0$ gives the magnitude and phase shown below.

B7. Normalized frequency response, 5000 samples

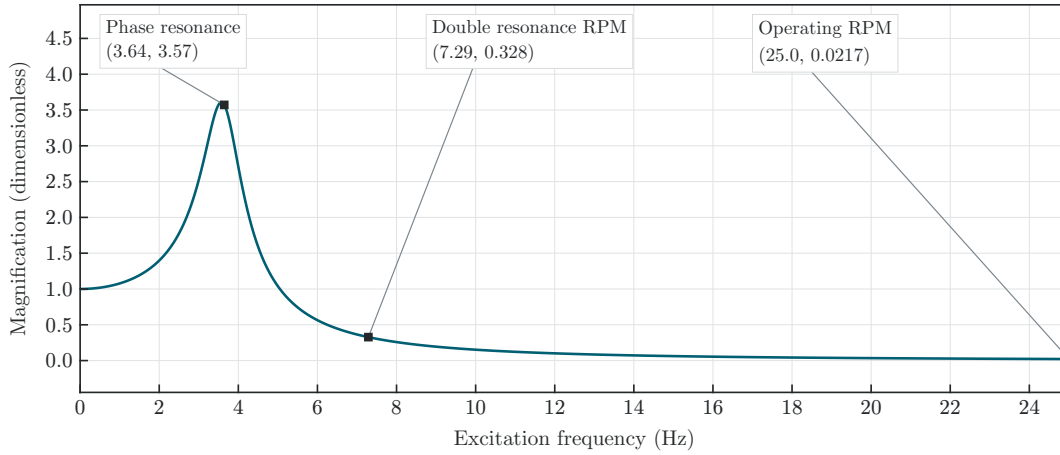


Figure 25: B.7.5.a. Dimensionless FRF magnitude from 0 to $f_0 = 25.0$ Hz.

B7. Response phase relative to force

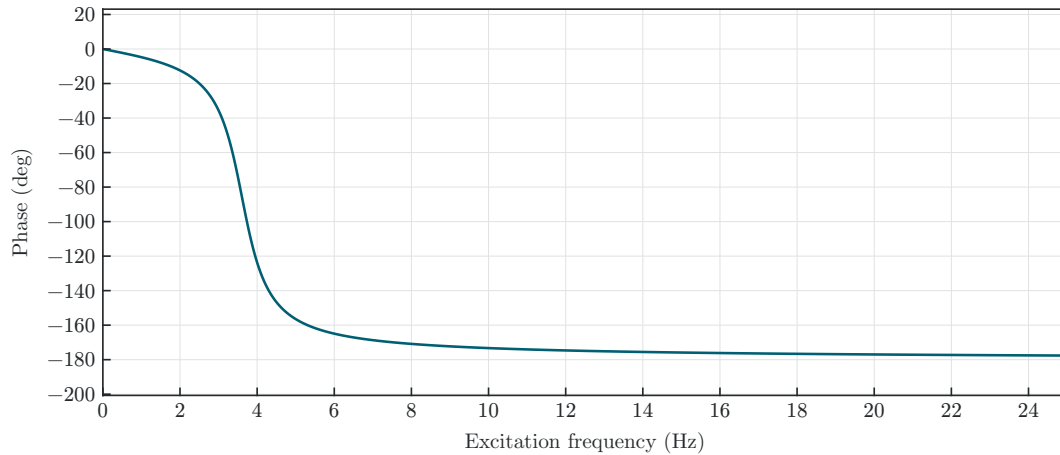


Figure 26: B.7.5.a. FRF phase from 0 to $f_0 = 25.0$ Hz.

(b) Requested FRF values. The phase-resonance speed is $N_{90} = 218.633$ rpm. Doubling that speed doubles the excitation frequency, so the second requested point is $2N_{90} = 437.265$ rpm and $f = 2f_n = 7.28775$ Hz.

Operating point	f (Hz)	N (rpm)	FRF
Phase resonance	3.64388	218.633	3.57143
Double phase-resonance speed	7.28775	437.265	0.32767
Operational speed	25.0000	1500.000	0.02169

(c) Operational vibration magnitude. At $f_0 = 25.0$ Hz, the frequency ratio is $p_0 = \omega_0/\omega_n = 6.86082$. Therefore,

$$|\hat{u}_0| = u_{\text{qst}} M(f_0) = (44.024 \mu\text{m})(0.0216868) = 0.954744 \mu\text{m}.$$

$$\boxed{|\hat{u}_0| = 0.954744 \mu\text{m}} \quad (\text{B.7.5.c})$$

(d) Run-up interpretation. When the motor starts, the displacement contains a transient because the mass begins away from the eventual forced steady state. As the motor accelerates, the excitation frequency passes through true resonance near $N_r = 214.305 \text{ rpm}$, where the steady-state magnification is largest. Phase resonance occurs slightly later at $N_{90} = 218.633 \text{ rpm}$, so the two speeds are close but not identical.

After the motor passes this region, the FRF magnitude decreases and the displacement becomes small at the operational speed. Under the supplied constant force amplitude, the predicted steady-state magnitude decreases from about $157 \mu\text{m}$ near phase resonance to $0.955 \mu\text{m}$ at 1500 rpm . The actual maximum during acceleration also depends on how quickly the motor crosses resonance. A rotating imbalance normally produces a force that varies with speed squared, so the fixed- $\hat{F}$ sweep describes the structural FRF and does not by itself predict the complete transient run-up.

MATLAB code. *Time response and FRF plots for B.7.*

```

01 L=.568; E=200e9; h=.003; b=.037; m=.52; zeta=.14;
02 Fhat=.012; N0=1500; Nosc=10; Nf=5000;
03 I=b*h^3/12; EI=E*I; k=3*EI/L^3; wn=sqrt(k/m);
04 wd=wn*sqrt(1-zeta^2); uqst=Fhat/k; U=uqst/(2*zeta);
05
06 t=linspace(0,Nosc*2*pi/wn,2000);
07 uss=U*sin(wn*t);
08 urest=uss-U*wn/wd*exp(-zeta*wn*t).*sin(wd*t);
09 plot(t,uss,'Color',[0 .427 .459],'LineWidth',1.5); hold on
10 plot(t,urest,'--','Color',[.776 .365 0],'LineWidth',1.5);
11 xlabel('Time (s)'); ylabel('Displacement (m)'); grid on
12
13 f=linspace(0,N0/60,Nf); p=2*pi*f/wn;
14 M=1./sqrt((1-p.^2).^2+(2*zeta*p).^2);
15 phi=-atan2(2*zeta*p,1-p.^2)*180/pi;
16 figure
17 plot(f,M,'Color',[0 .427 .459],'LineWidth',1.5);
18 figure
19 plot(f,phi,'Color',[0 .427 .459],'LineWidth',1.5);

```

B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact

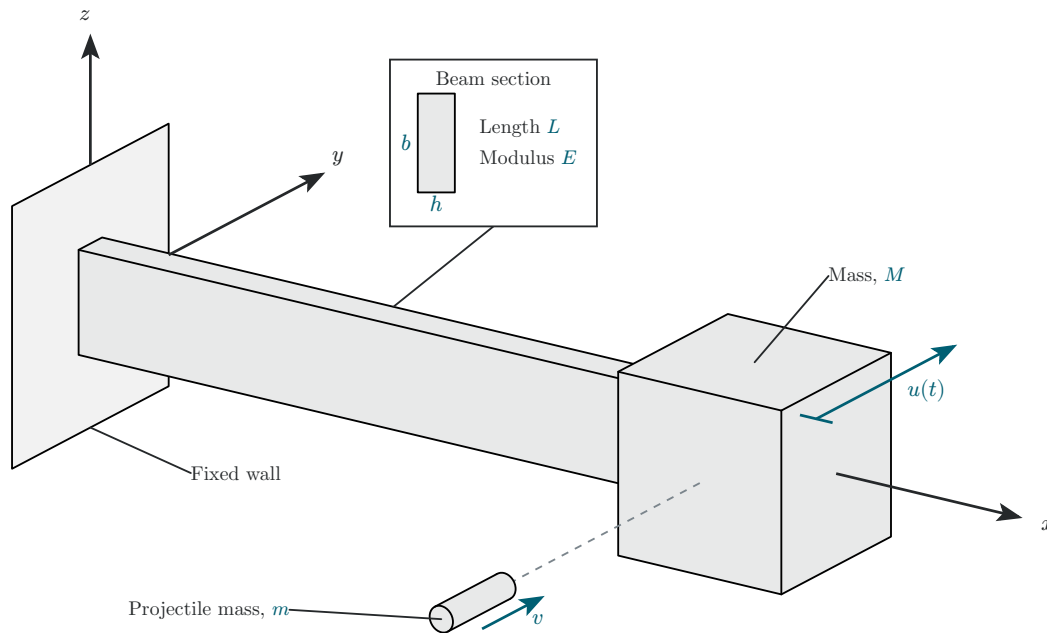


Figure 27: Impact excitation of a horizontal flex-beam vibration system.

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $M = 53 \text{ kg}$ attached at its tip as shown in Figure 27. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The beam is initially at rest. A small projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ .
- (7) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response calculated at item (7).

- (9) Discuss your results.

Solution to B.8

B.8.1. Input data. The projectile mass is denoted by m_p to distinguish it from the attached mass M . Since the projectile remains embedded after impact, the vibrating mass is $m_T = M + m_p$. I represent the beam by its equivalent tip stiffness.

Quantity	Value	Unit
L	0.568	m
E	200	GPa
h	3.00	mm
b	37.0	mm
M	53.0	kg
m_p	0.0180	kg
v_p	935	m/s
t_1	10.0	s
N_{osc}	10	oscillations

B.8.2. Second moment of area. The beam moves horizontally, but the bending axis is determined by the cross-section orientation. In the section shown, h is the bending depth and b is the dimension normal to that depth. Therefore,

$$I = \frac{bh^3}{12} = \frac{(0.037)(0.003)^3}{12} = 8.325 \times 10^{-11} \text{ m}^4 = 83.25 \text{ mm}^4.$$

$$\boxed{I = 8.325 \times 10^{-11} \text{ m}^4 \quad (83.25 \text{ mm}^4)} \quad (\text{B.8.2})$$

B.8.3. Cross-sectional stiffness. Multiplying the elastic modulus by the second moment of area gives

$$EI = (200 \times 10^9)(8.325 \times 10^{-11}) = 16.65 \text{ N m}^2.$$

$$\boxed{EI = 16.65 \text{ N m}^2} \quad (\text{B.8.3})$$

B.8.4. Flex-beam spring constant. The fixed-free beam tip deflection under a concentrated force gives the equivalent spring constant

$$k = \frac{3EI}{L^3} = \frac{3(16.65)}{(0.568)^3} = 272.578 \text{ N/m}.$$

$$\boxed{k = 272.578 \text{ N/m}} \quad (\text{B.8.4})$$

B.8.5. Initial velocity after impact. The collision duration is short compared with the vibration period, so the beam spring force contributes negligible impulse during the collision. Conservation of linear momentum therefore gives

$$m_p v_p + M(0) = (M + m_p) v_0.$$

Solving for the common velocity immediately after impact gives

$$v_0 = \frac{m_p v_p}{M + m_p} = \frac{(0.0180)(935)}{53.018} = 0.317439 \text{ m/s}.$$

$$\boxed{\dot{u}_0 = v_0 = 0.317439 \text{ m/s}} \quad (\text{B.8.5})$$

B.8.6. Natural frequency and period. After the collision, the projectile and attached mass move together. Because the beam is horizontal and damping is neglected, the free-vibration equation is

$$(M + m_p)\ddot{u} + ku = 0.$$

The natural frequency follows from the characteristic equation.

$$\omega_n = \sqrt{\frac{k}{M + m_p}} = \sqrt{\frac{272.578}{53.018}} = 2.26743 \text{ rad/s}.$$

The period is the time required for one complete cycle.

$$\tau = \frac{2\pi}{\omega_n} = \frac{2\pi}{2.26743} = 2.77106 \text{ s}.$$

$$\boxed{\omega_n = 2.26743 \text{ rad/s}, \quad \tau = 2.77106 \text{ s}} \quad (\text{B.8.6})$$

B.8.7. Displacement at $t_1 = 10 \text{ s}$. The beam displacement is continuous through the impact, so $u(0) = 0$. The initial velocity is $\dot{u}(0) = v_0$. Applying these conditions to the undamped free-vibration solution gives

$$u(t) = u(0) \cos(\omega_n t) + \frac{\dot{u}(0)}{\omega_n} \sin(\omega_n t) = \frac{v_0}{\omega_n} \sin(\omega_n t).$$

At $t_1 = 10 \text{ s}$,

$$u(t_1) = \frac{0.317439}{2.26743} \sin((2.26743)(10)) = -0.0883717 \text{ m.}$$

$$\boxed{u(10 \text{ s}) = -88.3717 \text{ mm}} \quad (\text{B.8.7})$$

B.8.8. Ten-oscillation response.

(a) **Largest response.** The response amplitude is

$$|u|_{\max} = \frac{v_0}{\omega_n} = \frac{0.317439}{2.26743} = 0.1399997 \text{ m.}$$

The first positive maximum occurs when $\omega_n t = \pi/2$, so

$$t_{\text{peak},1} = \frac{\pi}{2\omega_n} = 0.692766 \text{ s.}$$

$$\boxed{|u|_{\max} = 140.000 \text{ mm} \quad \text{at} \quad t_{\text{peak},1} = 0.692766 \text{ s}} \quad (\text{B.8.8.a})$$

(b) **Plot markers.** Ten oscillations span

$$t_{\text{end}} = N_{\text{osc}}\tau = 10(2.77106) = 27.7106 \text{ s.}$$

The plot marks the first positive maximum and the response at $t_1 = 10 \text{ s}$. Their coordinates are (0.693 s, 140.0 mm) and (10.00 s, -88.4 mm).

B8. Response after an embedded-projectile impact

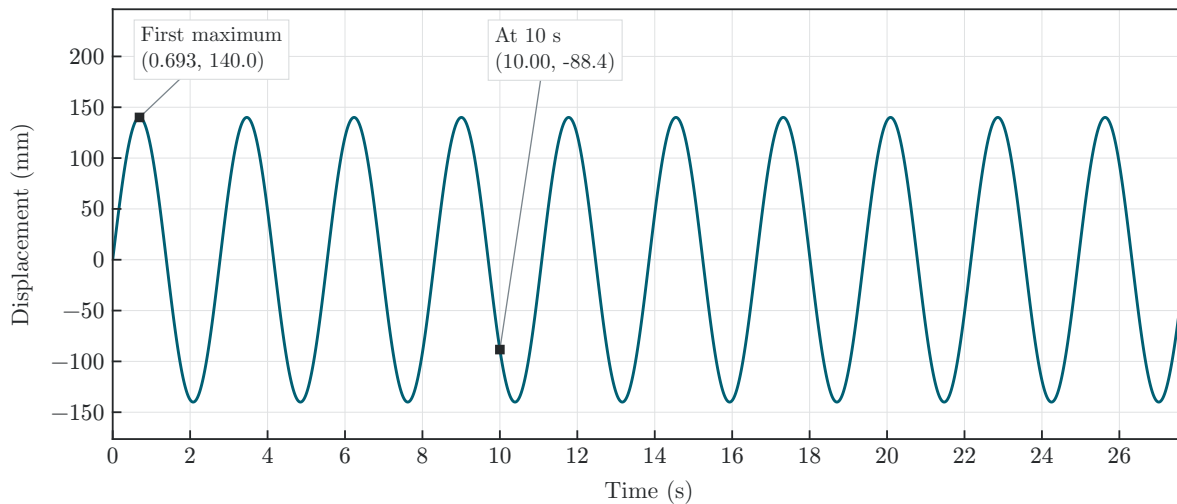


Figure 28: B.8.8.a-b. Undamped horizontal flex-beam response over $N_{\text{osc}} = 10$ cycles.

B.8.9. Discussion. The projectile transfers its momentum to the combined mass, which creates the initial velocity v_0 while leaving the displacement equal to zero. The undamped model then preserves the amplitude at approximately 140 mm. This is about 24.6% of the 568 mm span, so the result is large enough that small-deflection beam theory should be treated as a linear-model estimate.

The negative value at 10 s is caused by the phase of the sinusoidal response at that time; it does not indicate a change in amplitude. The calculation also omits distributed beam mass, damping, local impact deformation, and large-deflection geometric effects. Including those effects would change the natural frequency and reduce or otherwise modify the observed response.

MATLAB code. *Ten-oscillation response after the embedded-projectile impact.*

```
01 L=.568; E=200e9; h=.003; b=.037;
02 M=53; mp=.018; vp=935; t1=10; Nosc=10;
03 I=b*h^3/12; k=3*E*I/L^3; mt=M+mp;
04 v0=mp*vp/mt; wn=sqrt(k/mt); tau=2*pi/wn;
05 A=v0/wn; t=linspace(0,Nosc*tau,2000);
06 u=A*sin(wn*t); u1=A*sin(wn*t1);
07 plot(t,1000*u,'Color',[0 .427 .459],'LineWidth',1.5); hold on
08 plot(tau/4,1000*A,'ks',t1,1000*u1,'ks');
09 xlabel('Time (s)'); ylabel('Displacement (mm)'); grid on
```

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Vertical Spring-Mass Vibration System

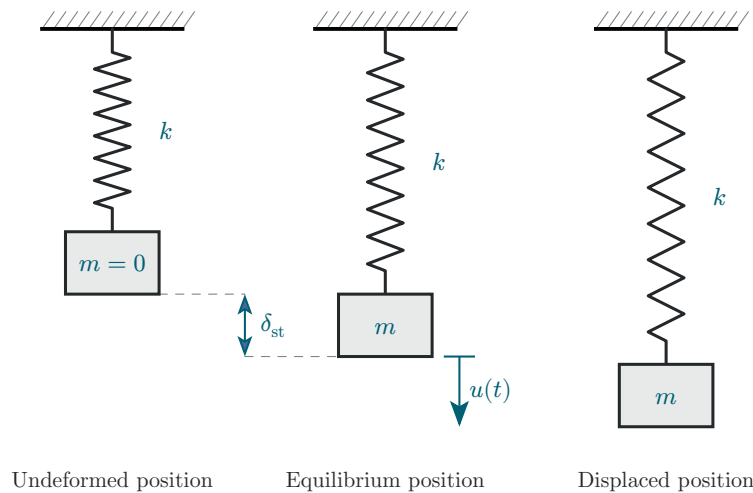


Figure 29: Vertical spring-mass vibration system.

Consider a vertical spring-mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 29.

Do the following.

- (1) Assume static equilibrium.
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (2) Assume vibration with time-varying vertical displacement $u(t)$.
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal spring-mass vibration system of Problem B.1.
- (3) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

Solution to C.1

C.1.1(a). Weight. I take the downward direction as positive and measure the coordinate y from the undeformed spring position. With this convention, the gravitational force is

$$W = mg.$$

$$\boxed{W = mg} \quad (\text{C.1.1.a})$$

C.1.1(b) and C.1.2(a). Free-body diagrams. The left panel shows the mass at static equilibrium. The right panel shows the mass displaced downward by $u(t)$ from that equilibrium. The spring forces are upward because the spring opposes the downward displacement.

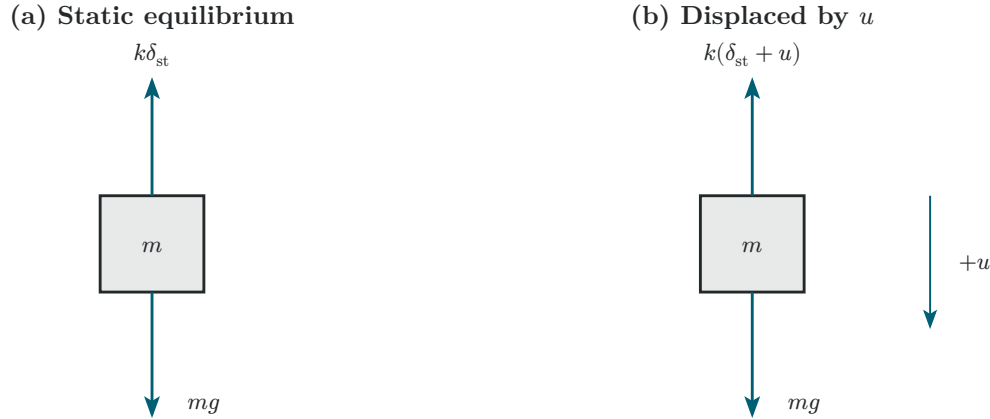


Figure 30: C.1.1.b and C.1.2.a. Free-body diagrams for static equilibrium and vertical vibration.

C.1.1(c). Static equilibrium equation. At equilibrium the acceleration is zero, so the downward weight and upward spring force balance. With downward positive,

$$\sum F_y = mg - k\delta_{\text{st}} = 0.$$

$$\boxed{mg - k\delta_{\text{st}} = 0} \quad (\text{C.1.1.c})$$

C.1.1(d). Static displacement. Rearranging the equilibrium equation gives

$$\delta_{\text{st}} = \frac{mg}{k}.$$

$$\boxed{\delta_{\text{st}} = \frac{mg}{k}} \quad (\text{C.1.1.d})$$

C.1.2(b). Equation of motion. Since $u(t)$ is measured from static equilibrium, the absolute downward position is

$$y(t) = \delta_{\text{st}} + u(t).$$

Because the spring deformation is $\delta_{\text{st}} + u(t)$, the net vertical force is

$$\sum F_y = mg - k(\delta_{\text{st}} + u(t)) = (mg - k\delta_{\text{st}}) - ku(t).$$

The first parenthesis is zero from static equilibrium. Newton's law therefore gives

$$m\ddot{u}(t) = \sum F_y = -ku(t),$$

which becomes the homogeneous free-vibration equation

$$\boxed{m\ddot{u}(t) + ku(t) = 0} \quad (\text{C.1.2.b})$$

C.1.2(c). Comparison with B.1. The equation is the same as the horizontal spring-mass equation because the constant gravitational force is canceled by the static spring force. Gravity changes the absolute equilibrium position through δ_{st} , but it does not appear in the incremental vibration equation. Dividing by m gives

$$\boxed{\ddot{u}(t) + \frac{k}{m}u(t) = 0, \quad \omega_n = \sqrt{\frac{k}{m}}} \quad (\text{C.1.2.c})$$

C.1.3. Arbitrary force field. Let $F(y, t)$ be the force magnitude and define its signed component along the downward coordinate by

$$Q(y, t) = F(y, t) \cos \theta(y, t).$$

The absolute equation of motion becomes

$$m\ddot{y} + ky = mg + Q(y, t).$$

For a time-independent field, let y_e be the equilibrium position, so that $ky_e = mg + Q(y_e)$. Setting $y = y_e + u$ and subtracting the equilibrium equation gives

$$m\ddot{u} + ku = Q(y_e + u) - Q(y_e).$$

Thus, a constant force component only shifts the equilibrium and disappears from the incremental equation. A position-dependent field changes the effective stiffness. For small displacement, linearization gives

$$Q(y_e + u) - Q(y_e) \approx Q'(y_e)u,$$

so the linearized equation is

$$\boxed{m\ddot{u} + [k - Q'(y_e)]u = 0} \quad (\text{C.1.3})$$

If the field varies with time, its time-varying component remains as an external forcing term. A force component perpendicular to the allowed motion is handled by the constraint in this one-coordinate model; otherwise, an additional coordinate and a coupled equation are required.

C.2 Analyze Vertical Flex-Beam Vibration

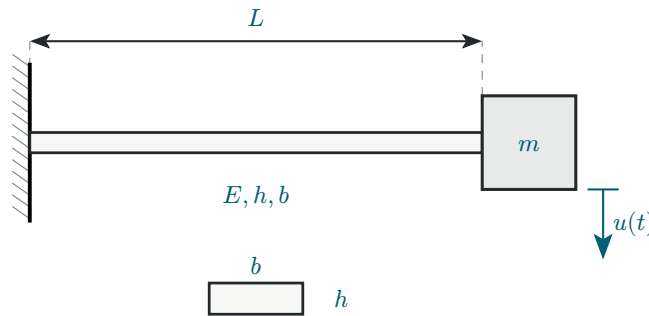


Figure 31: Vertical flex-beam vibrating system.

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 31. The beam cross-section properties are E , h , b .

Do the following.

- (1) Calculate the cross-sectional stiffness EI of the beam for vertical flexural deformation.
- (2) Calculate the spring constant k of the system.
- (3) Assume static equilibrium:
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (4) Assume vibration with time-varying vertical displacement $u(t)$ measured from the static equilibrium position:
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal flex-beam vibration system of Problem C.4.
- (5) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

Solution to C.2

C.2.1. Cross-sectional stiffness. The cross section is rectangular with width b and vertical bending depth h . Since the beam deflects vertically, the distance from the neutral axis is measured through h , so the second moment of area is

$$I = \frac{bh^3}{12}.$$

Multiplying by the elastic modulus gives the cross-sectional stiffness

$$EI = E \frac{bh^3}{12}.$$

$$\boxed{I = \frac{bh^3}{12}, \quad EI = \frac{Ebh^3}{12}} \quad (\text{C.2.1})$$

C.2.2. Spring constant. For a concentrated tip load P , the fixed-free beam tip deflection is

$$\Delta_{\max} = \frac{PL^3}{3EI}.$$

The equivalent tip spring constant is the load divided by the tip deflection. Therefore,

$$k = \frac{P}{\Delta_{\max}} = \frac{3EI}{L^3} = \frac{3Ebh^3}{12L^3}.$$

$$\boxed{k = \frac{3EI}{L^3}} \quad (\text{C.2.2})$$

C.2.3(a). Weight. The gravitational force acting on the tip mass is

$$W = mg.$$

$$\boxed{W = mg} \quad (\text{C.2.3.a})$$

C.2.3(b) and C.2.4(a). Free-body diagrams. The left panel shows the mass at static equilibrium, and the right panel shows the same mass displaced downward by $u(t)$ from equilibrium. The cantilever is represented by its equivalent vertical spring.

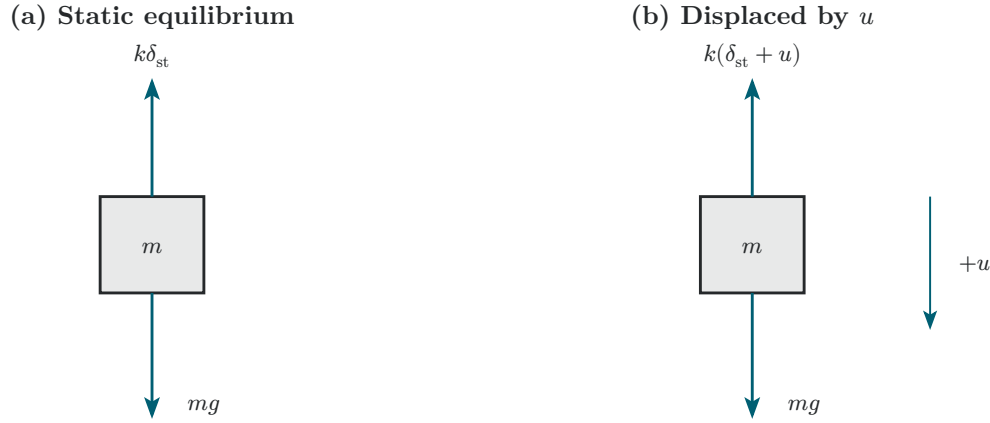


Figure 32: C.2.3.b and C.2.4.a. Free-body diagrams for static equilibrium and vertical flex-beam vibration.

C.2.3(c). Static equilibrium equation. With downward positive, the mass is stationary when the weight is balanced by the upward beam reaction. The equivalent spring force at the static deflection is $k\delta_{st}$, so

$$\sum F_y = mg - k\delta_{st} = 0.$$

$$\boxed{mg - k\delta_{st} = 0} \quad (\text{C.2.3.c})$$

C.2.3(d). Static displacement. Solving the equilibrium equation for the deflection gives

$$\delta_{st} = \frac{mg}{k} = \frac{mgL^3}{3EI}.$$

$$\boxed{\delta_{st} = \frac{mg}{k} = \frac{mgL^3}{3EI}} \quad (\text{C.2.3.d})$$

C.2.4(b). Equation of motion. Since the vibration displacement is measured from static equilibrium, the absolute downward position is

$$y(t) = \delta_{st} + u(t).$$

The beam deformation is then $\delta_{st} + u(t)$, which gives the net vertical force

$$\sum F_y = mg - k(\delta_{st} + u(t)) = (mg - k\delta_{st}) - ku(t).$$

The first parenthesis vanishes by static equilibrium. Since δ_{st} is constant, $\ddot{y} = \ddot{u}$, and Newton's law gives

$$m\ddot{u}(t) = \sum F_y = -ku(t).$$

Therefore, the vertical flex-beam equation of motion is

$$\boxed{m\ddot{u}(t) + ku(t) = 0} \quad (\text{C.2.4.b})$$

C.2.4(c). Comparison with C.4. The horizontal flex-beam model used in C.4 has the same lumped-mass equation because its cantilever is also replaced by the equivalent tip stiffness $k = 3EI/L^3$. After the displacement is measured from equilibrium, both systems reduce to

$$\ddot{u}(t) + \frac{k}{m}u(t) = 0, \quad \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{3EI}{mL^3}}.$$

The orientation affects which cross-sectional dimension is cubed in I , while gravity affects the vertical system's static offset. Neither changes the form of the incremental free-vibration equation.

$$\boxed{m\ddot{u}(t) + ku(t) = 0, \quad \omega_n = \sqrt{\frac{3EI}{mL^3}}} \quad (\text{C.2.4.c})$$

C.2.5. Arbitrary force field. Let $F(y, t)$ be the force magnitude and let $\theta(y, t)$ be its angle from the downward vertical. Its component along the beam's allowed motion is

$$Q(y, t) = F(y, t) \cos \theta(y, t).$$

The absolute vertical equation becomes

$$m\ddot{y} + ky = mg + Q(y, t).$$

For a time-independent field, let y_e satisfy $ky_e = mg + Q(y_e)$. Setting $y = y_e + u$ and subtracting the equilibrium equation gives

$$m\ddot{u} + ku = Q(y_e + u) - Q(y_e).$$

If the field is differentiable and the vibration is small, then

$$Q(y_e + u) - Q(y_e) \approx Q'(y_e)u,$$

so the linearized equation is

$$\boxed{m\ddot{u} + [k - Q'(y_e)]u = 0} \quad (\text{C.2.5})$$

A constant force component only shifts the equilibrium. A position-dependent component changes the effective stiffness, and a time-dependent component remains as an external forcing term. A perpendicular component is carried by the support only in the one-coordinate model; otherwise, the model must include additional coupled motion.

C.3 Analyze Horizontal Flex-Beam Vibration

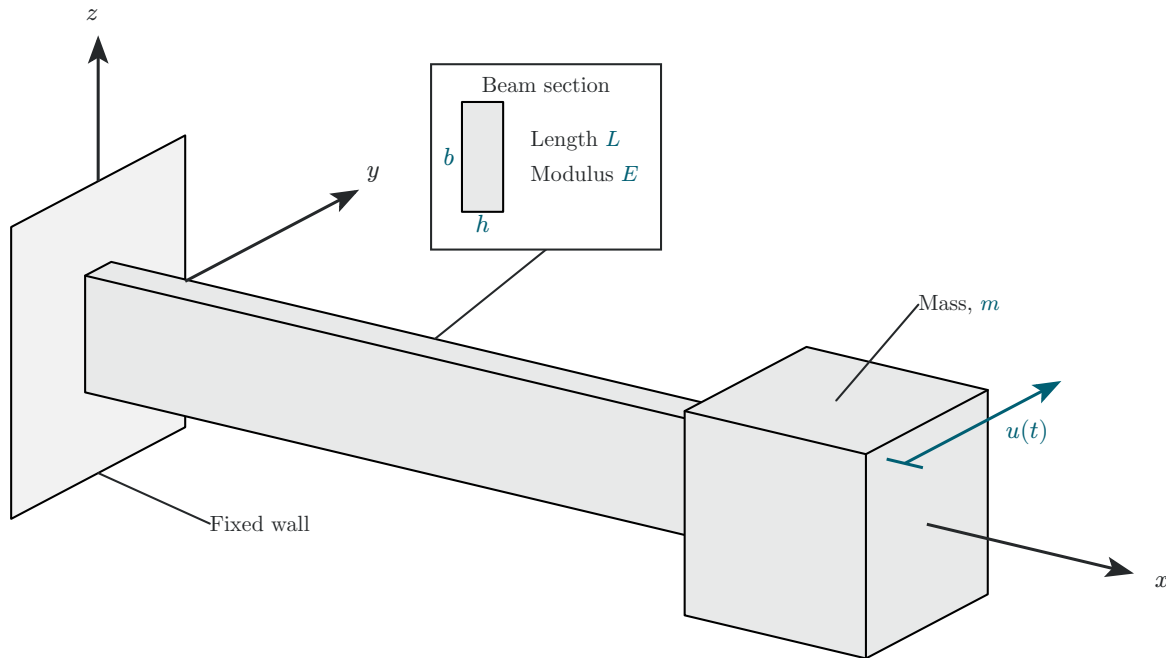


Figure 33: Horizontal flex-beam vibrating system.

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 33. The beam cross-section properties are E , h , b .

Do the following.

- (1) Assume vibration with horizontal time-varying displacement $u(t)$ as shown in Figure 33.
- (2) Calculate the cross-sectional stiffness EI of the beam for flexural deformation consistent with $u(t)$.
- (3) Calculate the flex-beam spring constant k .
- (4) Draw free-body diagram (FBD) including all the forces acting on the mass m .
- (5) Deduce equation of motion (EOM).
- (6) Compare the EOM deduced here with the EOM derived in Problem B.1.
- (7) Solve the initial value problem (IVP) and find the response for simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to C.3

C.3.1. Model and assumptions. I model the beam as a uniform, massless, linearly elastic cantilever with a lumped tip mass m . I assume small horizontal deflection, one horizontal degree of freedom, no damping, and no applied horizontal force. Since the vertical support reaction balances the weight, the horizontal vibration can be analyzed independently.

C.3.2. Cross-sectional stiffness. The beam bends in the horizontal direction shown in Figure 33. The cross-sectional dimension measured through the horizontal deflection direction is h , while b is normal to that direction. Therefore, the relevant second moment of area is

$$I = \frac{bh^3}{12}.$$

Multiplying by the elastic modulus gives

$$EI = E \frac{bh^3}{12}.$$

$$\boxed{I = \frac{bh^3}{12}, \quad EI = \frac{Ebh^3}{12}} \quad (\text{C.3.2})$$

C.3.3. Flex-beam spring constant. For a concentrated load P at the free end of a cantilever, the maximum tip deflection is

$$\Delta_{\max} = \frac{PL^3}{3EI}.$$

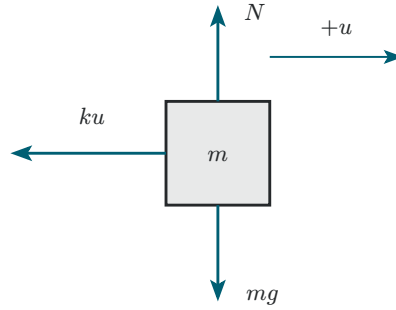
The equivalent tip spring constant is the load divided by the tip deflection. Hence,

$$k = \frac{P}{\Delta_{\max}} = \frac{3EI}{L^3} = \frac{3Ebh^3}{12L^3}.$$

$$\boxed{k = \frac{3EI}{L^3}} \quad (\text{C.3.3})$$

C.3.4. Free-body diagram. I take the horizontal displacement $u(t)$ as positive to the right. The beam's equivalent spring force therefore acts to the left with magnitude ku , while the vertical support reaction N balances the downward weight mg .

(a) Horizontal free vibration

**Figure 34:** C.3.4. Free-body diagram for horizontal flex-beam vibration.

C.3.5. Equation of motion. The vertical forces balance because the mass has no vertical acceleration,

$$\sum F_z = N - mg = 0.$$

In the horizontal direction, the spring reaction opposes the displacement, so

$$\sum F_x = -ku(t).$$

Applying Newton's law in the direction of motion gives

$$m\ddot{u}(t) = \sum F_x = -ku(t).$$

Rearranging produces the horizontal flex-beam equation of motion,

$$\boxed{m\ddot{u}(t) + ku(t) = 0} \quad (\text{C.3.5})$$

C.3.6. Comparison with B.1. The equation has the same form as the horizontal spring-mass equation in B.1. The difference is that the flex-beam stiffness is obtained from the cantilever relation $k = 3EI/L^3$, which gives

$$\ddot{u}(t) + \frac{k}{m}u(t) = 0, \quad \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{3EI}{mL^3}}.$$

Thus, the beam changes the value of k and ω_n , but not the form of the one-degree-of-freedom equation.

$$\boxed{m\ddot{u}(t) + ku(t) = 0, \quad \omega_n = \sqrt{\frac{3EI}{mL^3}}} \quad (\text{C.3.6})$$

C.3.7. Initial value problem. The undamped solution is written as

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t).$$

Evaluating the displacement at $t = 0$ gives $A = u_0$. Differentiating the response and evaluating the velocity at $t = 0$ gives $\dot{u}_0 = \omega_n B$, so $B = \dot{u}_0/\omega_n$. Substitution gives the response for simultaneous initial displacement and initial velocity,

$$u(t) = u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t), \quad \omega_n = \sqrt{\frac{3EI}{mL^3}} \quad (\text{C.3.7})$$

C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement

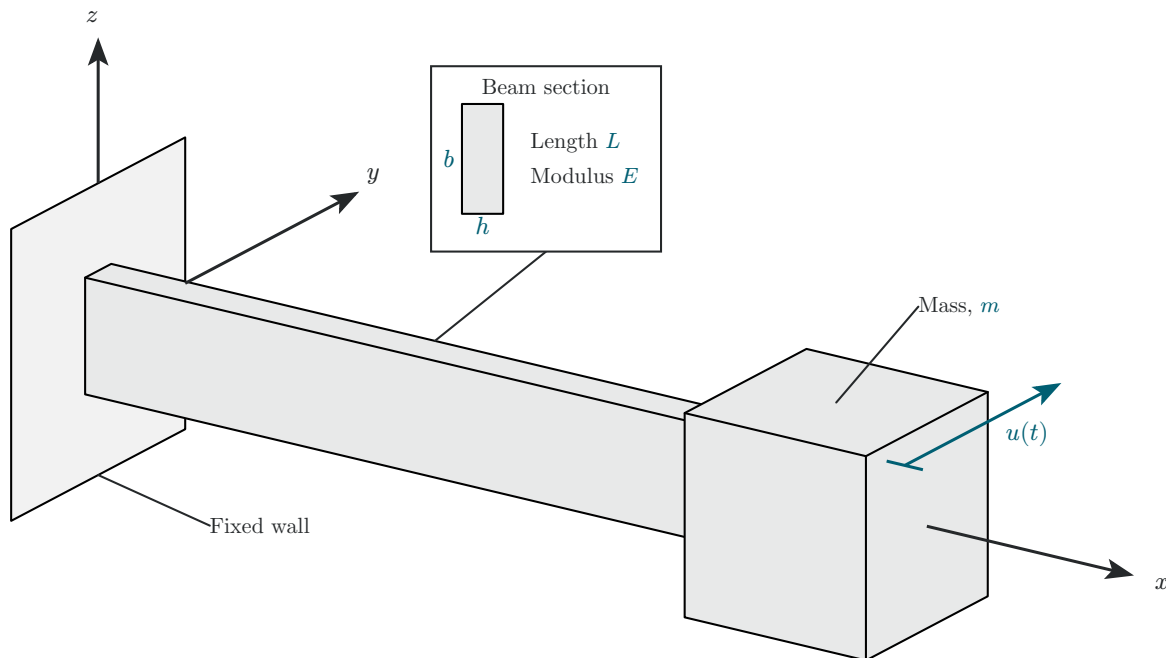


Figure 35: *Flex-beam vibrating system.*

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $m = 5.35 \text{ kg}$ attached at its tip as shown in Figure 35. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The tip mass is given an initial displacement $u_0 = 5.4 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n .
- (6) Calculate the natural frequency in Hz f_n .
- (7) Calculate the period of oscillation in sec τ .
- (8) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (9) Plot the response of the system for $N_{\text{osc}} = 25$ oscillations and use a data tip to find $u(t_1)$ at t_1 .
- (10) Discuss your results.

Solution to C.4

Using SI units, the given system data are

$$L = 0.568 \text{ m}, \quad m = 5.35 \text{ kg}, \quad E = 200 \times 10^9 \text{ Pa},$$

$$b = 0.037 \text{ m}, \quad h = 0.003 \text{ m}, \quad u_0 = 0.0054 \text{ m}, \quad v_0 = 0.$$

These values use SI units throughout, so the displacement response will first be calculated in metres.

$$\boxed{\begin{array}{l} L = 0.568 \text{ m}, \quad m = 5.35 \text{ kg}, \quad E = 200 \text{ GPa}, \\ b = 0.037 \text{ m}, \quad h = 0.003 \text{ m}, \quad u_0 = 5.4 \text{ mm}, \quad v_0 = 0 \end{array}} \quad (\text{C.4.1})$$

For the rectangular flex-beam cross-section, the second moment of area about the bending axis is

$$I = \frac{bh^3}{12} = \frac{(0.037)(0.003)^3}{12} = 8.325 \times 10^{-11} \text{ m}^4.$$

Because $1 \text{ m}^4 = 10^{12} \text{ mm}^4$, this is 83.25 mm^4 .

$$\boxed{I = 8.325 \times 10^{-11} \text{ m}^4 = 83.25 \text{ mm}^4} \quad (\text{C.4.2})$$

The flexural rigidity follows by multiplying the second moment of area by Young's modulus. Therefore,

$$EI = (200 \times 10^9)(8.325 \times 10^{-11}) = 16.65 \text{ N m}^2.$$

$$\boxed{EI = 16.65 \text{ N m}^2} \quad (\text{C.4.3})$$

A cantilever beam with a concentrated tip load has the end displacement $\Delta = PL^3/(3EI)$. Hence its equivalent transverse stiffness is

$$k = \frac{P}{\Delta} = \frac{3EI}{L^3} = \frac{3(16.65)}{(0.568)^3} = 272.5778 \text{ N/m}.$$

$$\boxed{k = 272.58 \text{ N/m}} \quad (\text{C.4.4})$$

With displacement measured from the undeformed reference used for this horizontal system, the undamped equation of motion is $m\ddot{u} + ku = 0$. Dividing by m gives $\ddot{u} + \omega_n^2 u = 0$, where $\omega_n^2 = k/m$. Thus,

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{272.5778}{5.35}} = 7.1379 \text{ rad/s}.$$

$$\boxed{\omega_n = 7.138 \text{ rad/s}} \quad (\text{C.4.5})$$

The cyclic natural frequency is the angular natural frequency divided by 2π , so

$$f_n = \frac{\omega_n}{2\pi} = \frac{7.1379}{2\pi} = 1.1360 \text{ Hz.}$$

$$\boxed{f_n = 1.136 \text{ Hz}} \quad (\text{C.4.6})$$

The period is the time for one complete oscillation. Therefore,

$$\tau = \frac{1}{f_n} = \frac{2\pi}{\omega_n} = 0.88026 \text{ s.}$$

$$\boxed{\tau = 0.8803 \text{ s}} \quad (\text{C.4.7})$$

The general solution of the undamped equation is

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t).$$

At $t = 0$, the initial displacement gives $A = u_0$, while the initial velocity gives $B = v_0/\omega_n$. Since $v_0 = 0$, the response reduces to

$$u(t) = u_0 \cos(\omega_n t) = 0.0054 \cos(7.1379t) \text{ m.}$$

Evaluating this expression at $t_1 = 10 \text{ s}$ gives

$$u(10) = 0.0054 \cos(7.1379(10)) = -0.00344903 \text{ m} = -3.44903 \text{ mm.}$$

$$\boxed{u(10 \text{ s}) = -3.449 \text{ mm}} \quad (\text{C.4.8})$$

For $N_{\text{osc}} = 25$, the plotting interval ends at

$$t_{\text{end}} = N_{\text{osc}}\tau = 25(0.88026) = 22.0065 \text{ s.}$$

The plotted response is shown below. The square marker identifies the calculated value at $t_1 = 10 \text{ s}$, and the same value can be read with a data tip.

C4. Horizontal cantilever displacement release

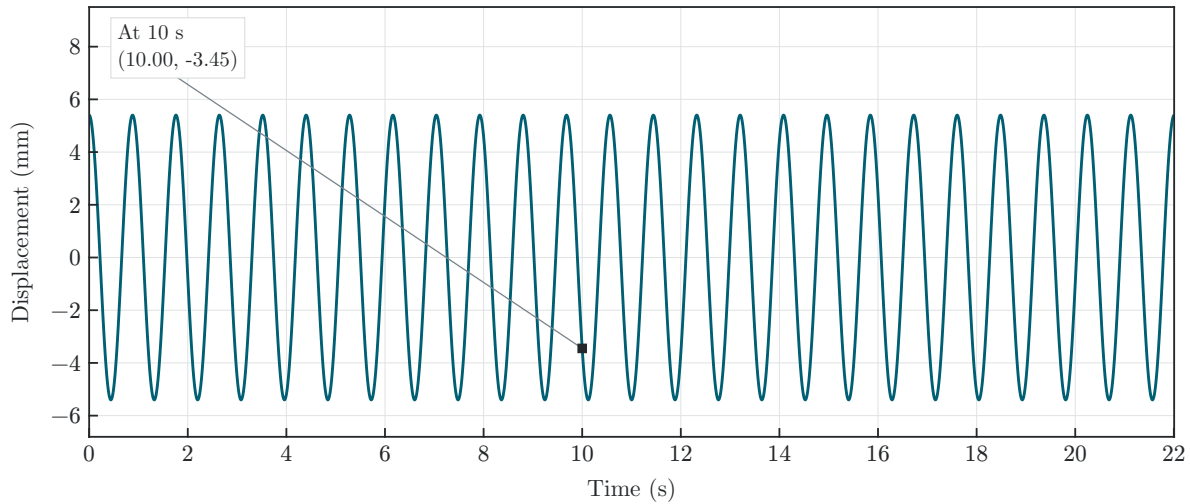


Figure 36: C.4. Twenty-five oscillations with the ten-second response marked.

MATLAB code. C.4 response plot.

```

01 L=.568; E=200e9; h=.003; b=.037; m=5.35;
02 u0=.0054; v0=0; t1=10; Nosc=25;
03 I=b*h^3/12; EI=E*I; k=3*EI/L^3;
04 wn=sqrt(k/m); fn=wn/(2*pi); tau=2*pi/wn;
05 t=linspace(0,Nosc*tau,2500);
06 u=u0*cos(wn*t)+v0/wn*sin(wn*t);
07 u1=u0*cos(wn*t1)+v0/wn*sin(wn*t1);
08 plot(t,1000*u,'Color',[0 .427 .459],'LineWidth',1.5);
09 hold on; plot(t1,1000*u1,'ks'); grid on;
10 xlabel('Time (s)'); ylabel('Displacement (mm)');

```

$$\begin{aligned}
 t_{\text{end}} &= 22.0065 \text{ s} \quad \text{for } N_{\text{osc}} = 25, \\
 u_{\text{tip}}(10 \text{ s}) &= -3.449 \text{ mm}
 \end{aligned}
 \tag{C.4.9}$$

The computed displacement and the data-tip value agree because both use the same undamped response. The displacement changes sign as the mass passes through equilibrium, but its maximum absolute value remains $u_0 = 5.4 \text{ mm}$. Since no damping or external force is included, the oscillation continues with constant amplitude and period.

$$\max |u(t)| = 5.4 \text{ mm}, \quad \tau = 0.8803 \text{ s with constant amplitude}
 \tag{C.4.10}$$

C.5 Analyze Spring-Mass Forced Vibration

Consider a spring-mass vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 37.

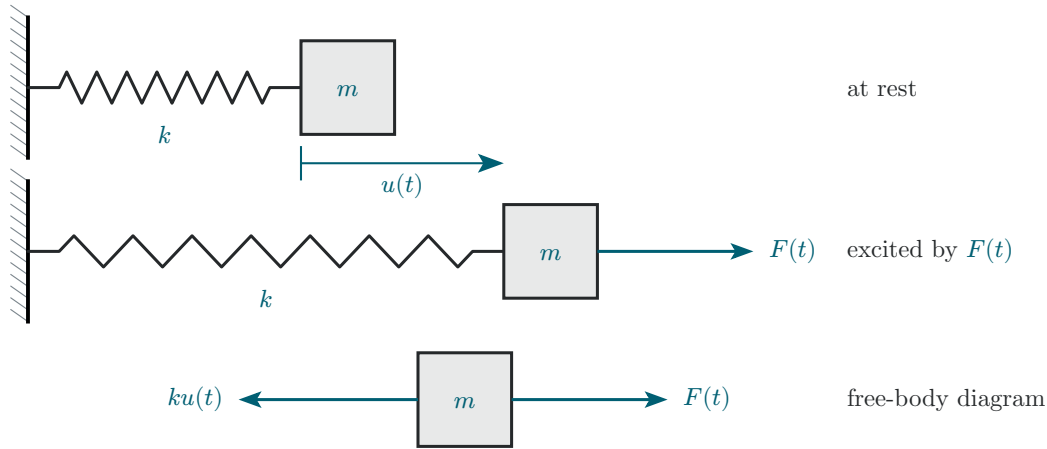


Figure 37: Spring-mass system excited by force $F(t)$.

General Analysis of Spring-Mass Forced Vibration

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) and express it as a nonhomogeneous ODE in terms of m , k , $F(t)$.
- (5) Write the ODE general solution as the superposition of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (6) Recall the complementary solution $u_c(t)$ as the solution of the homogeneous equation derived in Problem B.1 and write it in terms of trigonometric functions multiplied by constants A and B .

Spring-Mass Response to a Suddenly Applied Load

- (7) Assume the loading $F(t)$ is a suddenly applied load of magnitude F_0 .
- (8) Find the particular solution for this case of a suddenly applied load.
- (9) Write the complete solution as the sum of $u_c(t)$ from (6) and $u_p(t)$ from (8).
- (10) Assume “at rest” initial conditions to find the constants A and B and write the complete solution $u(t)$ as a function of F_0 , k , and $\cos \omega_n t$.
- (11) Use (10) to find the largest displacement $u_{\max}$ and compare it with the static displacement u_{st} .

Solution to C.5

I assume a single-degree-of-freedom mass-spring model with a rigid mass, a massless linear spring, small displacement, and no damping. The applied force acts in the positive x direction, while the spring force opposes the displacement.

$$\boxed{\text{Single degree of freedom, linear spring, massless spring, rigid mass, small displacement, no damping, and prescribed force } F(t)} \quad (\text{C.5.1})$$

For the mass shown in the free-body diagram, the applied force acts to the right and the spring reaction acts to the left. The force sum in the positive x direction is therefore

$$\sum F_x = F(t) - ku(t).$$

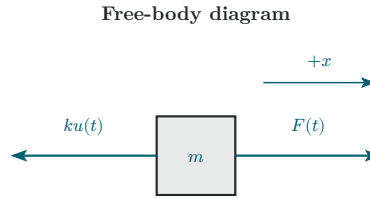


Figure 38: C.5. Free-body diagram for the forced spring-mass system.

$$\boxed{\sum F_x = F(t) - ku(t)} \quad (\text{C.5.2})$$

Newton's law of motion relates this force sum to the mass acceleration. Thus,

$$\sum F_x = m\ddot{u}(t) = F(t) - ku(t).$$

$$\boxed{\sum F_x = m\ddot{u}(t)} \quad (\text{C.5.3})$$

Rearranging the force balance gives the equation of motion,

$$m\ddot{u}(t) + ku(t) = F(t).$$

Dividing by m and using $\omega_n^2 = k/m$ gives the equivalent normalized form

$$\ddot{u}(t) + \omega_n^2 u(t) = \frac{F(t)}{m}, \quad \omega_n = \sqrt{\frac{k}{m}}.$$

$$\boxed{m\ddot{u}(t) + ku(t) = F(t), \quad \omega_n = \sqrt{\frac{k}{m}}} \quad (\text{C.5.4})$$

The right-hand side is nonzero, so the equation is nonhomogeneous. Its general response is the sum of a complementary solution and a particular solution,

$$u(t) = u_c(t) + u_p(t),$$

where $u_c(t)$ satisfies the homogeneous equation and $u_p(t)$ satisfies the forced equation.

$$\boxed{u(t) = u_c(t) + u_p(t)} \quad (\text{C.5.5})$$

Setting the force to zero gives the homogeneous equation $m\ddot{u}_c + ku_c = 0$. Its undamped solution is

$$u_c(t) = A \cos(\omega_n t) + B \sin(\omega_n t).$$

$$\boxed{u_c(t) = A \cos(\omega_n t) + B \sin(\omega_n t)} \quad (\text{C.5.6})$$

For the suddenly applied load, the force is switched on at $t = 0$ and then held constant. Therefore, for $t > 0$,

$$F(t) = F_0, \quad m\ddot{u} + ku = F_0.$$

$$\boxed{F(t) = F_0 \text{ for } t > 0} \quad (\text{C.5.7})$$

Because the forcing function is constant, the particular solution can be taken as $u_p = C$. Its derivatives vanish, so substitution into the nonhomogeneous equation gives

$$m(0) + kC = F_0.$$

Solving for C gives the particular solution

$$u_p(t) = \frac{F_0}{k}.$$

$$\boxed{u_p(t) = \frac{F_0}{k}} \quad (\text{C.5.8})$$

Adding the complementary and particular parts gives the complete response,

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) + \frac{F_0}{k}.$$

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) + \frac{F_0}{k} \quad (\text{C.5.9})$$

The mass starts at the unloaded position and is initially at rest. Hence,

$$u(0) = 0, \quad \dot{u}(0) = 0.$$

Evaluating the displacement condition gives

$$0 = A + \frac{F_0}{k}, \quad A = -\frac{F_0}{k}.$$

Differentiating the complete response and applying the velocity condition gives

$$0 = \omega_n B, \quad B = 0.$$

Substituting these constants into the complete response gives

$$u(t) = \frac{F_0}{k} [1 - \cos(\omega_n t)], \quad t > 0.$$

$$A = -\frac{F_0}{k}, \quad B = 0, \quad u(t) = \frac{F_0}{k} [1 - \cos(\omega_n t)] \quad (\text{C.5.10})$$

The static displacement is obtained by setting the acceleration to zero and allowing the force to act without vibration. Thus,

$$u_{\text{st}} = \frac{F_0}{k}.$$

The dynamic response is largest when $\cos(\omega_n t) = -1$, which first occurs at $t = \pi/\omega_n$. At that instant,

$$u_{\text{max}} = \frac{F_0}{k} [1 - (-1)] = \frac{2F_0}{k} = 2u_{\text{st}}.$$

Therefore, the dynamic amplification factor is

$$\frac{u_{\text{max}}}{u_{\text{st}}} = 2.$$

$$u_{\text{st}} = \frac{F_0}{k}, \quad u_{\text{max}} = u_{\text{dyn}} = \frac{2F_0}{k} = 2u_{\text{st}}, \quad \text{DAF} = 2 \quad (\text{C.5.11})$$

The suddenly applied force produces twice the static displacement because the mass has not yet moved when the load is switched on. The mass passes through the static equilibrium position with nonzero velocity, reaches the first turning point at π/ω_n , and then returns. With no damping, this oscillation continues with constant amplitude about the static displacement.

C.6 Calculate Cantilever Response to Suddenly Applied Weight

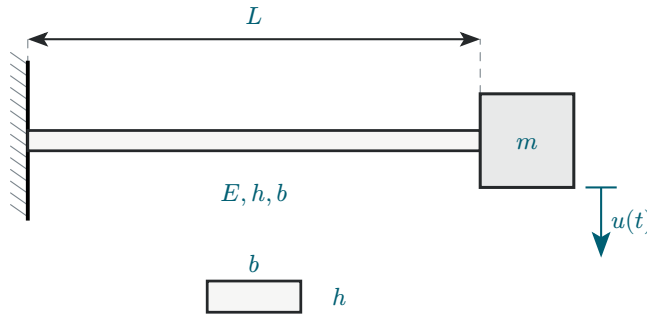


Figure 39: Cantilever beam with tip mass.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L = 568 \text{ mm}$ and a mass $m = 5.35 \text{ kg}$ that has to be attached to its tip as shown in Figure 39. The beam is made of steel with elastic modulus $E = 200 \text{ GPa}$, yield stress $Y = 710 \text{ MPa}$, and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$.

Do the following.

- (1) Display input data.
- (2) Calculate the cross-section second moment of area I , the flexural stiffness EI , the spring constant k , the natural frequency ω_n rad/s, f_n Hz, and period of oscillation τ of the system.
- (3) Calculate the weight F_0 of the mass m .
- (4) Static case. Assume the mass is attached to the undeformed beam and then released gradually without inducing vibration. Calculate the static displacement u_{st} .
- (5) Dynamic case. Assume the mass is attached to the undeformed beam and then released suddenly, inducing vibration.
 - (a) Calculate the maximum dynamic displacement u_{dyn} .
 - (b) Plot the response $u(t)$ measured from the undeformed position for $N_{\text{osc}} = 10$ oscillations.
 - (c) Place a data tip on the plot to identify the maximum displacement and compare it with the result of item (5)(a).
- (6) Graduate work. Calculate the maximum stress at the root of the beam and the safety factor $\text{SF} = Y/\sigma$, and comment on your results for the following two cases.
 - (a) Static-case stress σ_{st} .
 - (b) Dynamic-case stress σ_{dyn} .
- (7) Discuss your results.

Solution to C.6

Using SI units, the given parameters are

$$L = 0.568 \text{ m}, \quad m = 5.35 \text{ kg}, \quad E = 200 \times 10^9 \text{ Pa}, \quad h = 0.003 \text{ m}, \quad b = 0.037 \text{ m},$$

$$g = 9.81 \text{ m/s}^2, \quad Y = 710 \times 10^6 \text{ Pa}.$$

The mass is attached at the undeformed tip and is initially at rest.

$L = 0.568 \text{ m}, \quad m = 5.35 \text{ kg}, \quad E = 200 \text{ GPa},$ $h = 3 \text{ mm}, \quad b = 37 \text{ mm}, \quad g = 9.81 \text{ m/s}^2, \quad Y = 710 \text{ MPa}$	(C.6.1)
---	---------

The beam bends vertically, so the cross-section dimension h lies in the bending direction. For the rectangular cross-section, the second moment of area is

$$I = \frac{bh^3}{12} = \frac{(0.037)(0.003)^3}{12} = 8.325 \times 10^{-11} \text{ m}^4 = 83.25 \text{ mm}^4.$$

The flexural rigidity is then

$$EI = (200 \times 10^9)(8.325 \times 10^{-11}) = 16.65 \text{ N m}^2.$$

For a cantilever with a concentrated tip load, the end displacement is $\Delta = PL^3/(3EI)$. Therefore, the equivalent spring constant is

$$k = \frac{P}{\Delta} = \frac{3EI}{L^3} = \frac{3(16.65)}{(0.568)^3} = 272.5778 \text{ N/m}.$$

The natural frequency and period follow from

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{272.5778}{5.35}} = 7.1379 \text{ rad/s},$$

$$f_n = \frac{\omega_n}{2\pi} = 1.1360 \text{ Hz}, \quad \tau = \frac{2\pi}{\omega_n} = 0.88026 \text{ s}.$$

$I = 83.25 \text{ mm}^4, \quad EI = 16.65 \text{ N m}^2, \quad k = 272.58 \text{ N/m},$ $\omega_n = 7.138 \text{ rad/s}, \quad f_n = 1.136 \text{ Hz}, \quad \tau = 0.8803 \text{ s}$	(C.6.2)
---	---------

The applied weight is

$$F_0 = mg = (5.35)(9.81) = 52.4835 \text{ N}.$$

$F_0 = 52.48 \text{ N}$	(C.6.3)
-------------------------	---------

For the gradual-release case, the mass reaches equilibrium without oscillating. Setting the acceleration to zero gives

$$ku_{\text{st}} = F_0, \quad u_{\text{st}} = \frac{F_0}{k} = \frac{52.4835}{272.5778} = 0.192545 \text{ m.}$$

Thus, the static tip displacement is 192.545 mm.

$$\boxed{u_{\text{st}} = 192.55 \text{ mm}} \quad (\text{C.6.4})$$

For the sudden-release case, the load is applied at $t = 0$ while the mass is still at the undeformed position. The equation of motion for $t > 0$ is

$$m\ddot{u} + ku = F_0.$$

The at-rest initial conditions are

$$u(0) = 0, \quad \dot{u}(0) = 0.$$

Using the constant particular solution $u_p = F_0/k$ and the complementary solution $u_c = A \cos(\omega_n t) + B \sin(\omega_n t)$, these initial conditions give $A = -F_0/k$ and $B = 0$. Therefore, the displacement measured from the undeformed position is

$$u(t) = \frac{F_0}{k} [1 - \cos(\omega_n t)] = 0.192545 [1 - \cos(7.1379t)] \text{ m.}$$

The largest value occurs when $\cos(\omega_n t) = -1$, which first occurs at

$$t_{\text{max}} = \frac{\pi}{\omega_n} = 0.44013 \text{ s.}$$

Consequently,

$$u_{\text{dyn}} = u_{\text{max}} = \frac{2F_0}{k} = 2u_{\text{st}} = 0.385090 \text{ m.}$$

In millimetres, this is 385.090 mm.

$$\boxed{u_{\text{dyn}} = u_{\text{max}} = 385.09 \text{ mm} \quad \text{at} \quad t_{\text{max}} = 0.4401 \text{ s}} \quad (\text{C.6.5.a})$$

For $N_{\text{osc}} = 10$, the plotting interval is

$$t_{\text{end}} = N_{\text{osc}}\tau = 10(0.88026) = 8.8026 \text{ s.}$$

The response plot below is measured from the undeformed beam position. The dashed line is the static displacement, and the square marker identifies the first dynamic maximum.

C6. Ideal linear response to suddenly applied weight

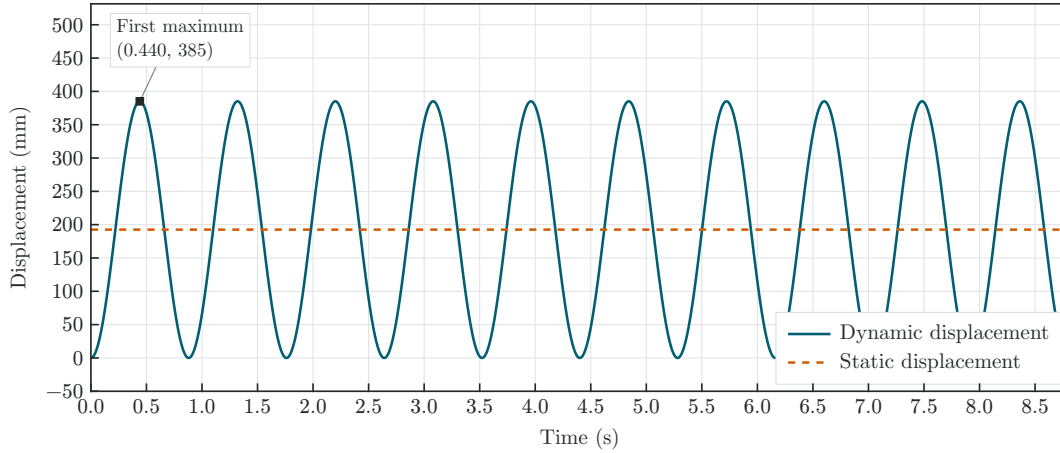


Figure 40: C.6. Ten-cycle response measured from the undeformed position.

$$N_{\text{osc}} = 10, \quad t_{\text{end}} = 8.8026 \text{ s} \quad (\text{C.6.5.b})$$

The data tip at the first maximum reads approximately

$$(t_{\text{max}}, u_{\text{max}}) = (0.440 \text{ s}, 385 \text{ mm}).$$

This agrees with the analytical value in item (5)(a) to the precision shown.

$$u_{\text{tip}}(0.440 \text{ s}) = 385 \text{ mm} \approx u_{\text{dyn}} \quad (\text{C.6.5.c})$$

MATLAB code. C.6 response plot.

```

01 L=.568; E=200e9; h=.003; b=.037; m=5.35; g=9.81;
02 Nosc=10; I=b*h^3/12; k=3*E*I/L^3; wn=sqrt(k/m);
03 F0=m*g; ust=F0/k; tau=2*pi/wn;
04 t=linspace(0,Nosc*tau,2000); u=ust*(1-cos(wn*t));
05 tmax=pi/wn; umax=2*ust;
06 plot(t,1000*u,'Color',[0 .427 .459],'LineWidth',1.5);
07 hold on; plot(t,1000*(ust+0*t),'--','Color',[.776 .365 0],'LineWidth',1.2);
08 plot(tmax,1000*umax,'ks'); grid on;
09 xlabel('Time (s)'); ylabel('Displacement (mm)');

```

For the static case, the root bending moment is the applied weight multiplied by the beam length,

$$M_{\text{st}} = F_0 L = (52.4835)(0.568) = 29.8106 \text{ N m}.$$

The outer-fibre distance is $c = h/2 = 0.0015 \text{ m}$. Using the flexure relation $\sigma = Mc/I$, the static root stress is

$$\sigma_{\text{st}} = \frac{M_{\text{st}} c}{I} = \frac{(29.8106)(0.0015)}{8.325 \times 10^{-11}} = 5.3713 \times 10^8 \text{ Pa} = 537.13 \text{ MPa}.$$

The corresponding safety factor is

$$SF_{st} = \frac{Y}{\sigma_{st}} = \frac{710}{537.13} = 1.322.$$

$$\boxed{\sigma_{st} = 537.13 \text{ MPa}, \quad SF_{st} = 1.322} \quad (\text{C.6.6.a})$$

At the dynamic maximum, the beam restoring force is

$$P_{dyn} = ku_{dyn} = 2F_0.$$

Therefore, the dynamic root moment is twice the static root moment,

$$M_{dyn} = P_{dyn}L = 2F_0L = 59.6213 \text{ N m}.$$

The maximum dynamic stress and its safety factor are

$$\sigma_{dyn} = \frac{M_{dyn}c}{I} = 1.0743 \times 10^9 \text{ Pa} = 1074.26 \text{ MPa},$$

$$SF_{dyn} = \frac{Y}{\sigma_{dyn}} = \frac{710}{1074.26} = 0.661.$$

$$\boxed{\sigma_{dyn} = 1074.26 \text{ MPa}, \quad SF_{dyn} = 0.661} \quad (\text{C.6.6.b})$$

The static safety factor is slightly greater than one, so the ideal static calculation remains below the specified yield stress. The sudden release doubles the tip displacement, root moment, and bending stress, which reduces the safety factor below one. The linear model therefore predicts yielding in the dynamic case. The predicted dynamic displacement is also $0.678L$, so large-deflection geometry and material nonlinearity would become important before the ideal linear response could be realized.

$$\boxed{SF_{dyn} < 1 \Rightarrow \text{yielding is predicted in the dynamic case}} \quad (\text{C.6.7})$$

C.7 Analyze Flex-Beam Pendulum

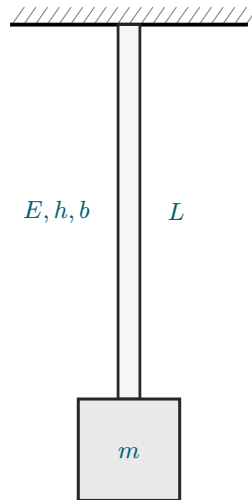


Figure 41: *Flex-beam pendulum.*

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 41. The beam is fixed into the ceiling and is hanging downward along the z axis. The mass may oscillate sideways in either x or y directions. The beam cross-section properties are E, h, b .

Do the following.

- (1) Calculate the minor and major second moments of area I_1 and I_2 .
- (2) Assume a horizontal displacement $u(t)$ which produces flexure about the minor I_1 .
- (3) Draw free-body diagram (FBD).
- (4) Write Newton's law of motion (NLM) equation.
- (5) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (6) Define the natural frequency $\omega_{n,\text{FBP}}$ of the flex-beam pendulum.
- (7) Deduce the governing equation.
- (8) Compare your results with those of Problem B.1 and write the general solution.
- (9) Solve the initial value problem (IVP), i.e., deduce the response for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.
- (10) Graduate work. Explain how the solution would change if flexure were about I_2 .

Solution to C.7

I model the uniform rectangular beam, with $b > h$, as a massless linear-elastic cantilever and the tip mass as a rigid point mass. I measure $u(t)$ horizontally from the undeflected vertical position and take positive u in the $+x$ direction.

$$I_1 = \frac{bh^3}{12}, \quad I_2 = \frac{hb^3}{12}.$$

Because h is the smaller cross-section dimension, bending about the x axis uses the minor second moment I_1 , while bending about the y axis uses the major second moment I_2 . Therefore,

$$\boxed{I_1 = \frac{bh^3}{12}, \quad I_2 = \frac{hb^3}{12}} \quad (\text{C.7.1})$$

For the prescribed motion, the beam flexes about I_1 . I represent it as an equivalent cantilever spring with tip displacement u . Under a transverse tip force P , the static cantilever response is

$$u = \frac{PL^3}{3EI_1}, \quad \theta = \frac{PL^2}{2EI_1},$$

where θ is the small tip slope. Eliminating P gives

$$k_1 = \frac{P}{u} = \frac{3EI_1}{L^3}, \quad \theta = \frac{3}{2L}u.$$

Thus, the elastic restoring force is represented by k_1u , while the tip slope needed for the gravity term is $3u/(2L)$.

$$\boxed{k_1 = \frac{3EI_1}{L^3}, \quad \theta = \frac{3u}{2L}} \quad (\text{C.7.2})$$

The free-body diagram contains the elastic restoring force k_1u , the horizontal component of the weight $mg \sin \theta$, and the positive displacement direction. The left panel shows the cantilever bending shape and defines the tip slope. The right panel reduces the motion to the tip mass and the two restoring forces.

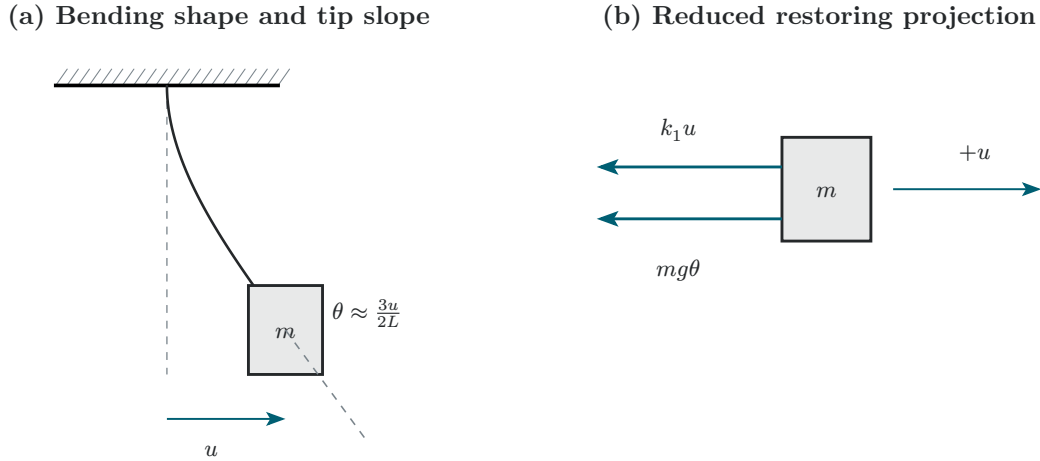


Figure 42: C.7.3. Flex-beam geometry and free-body diagram.

Taking $+x$ in the direction of u , both restoring forces act in the $-x$ direction. The force balance is therefore

$$\sum F_x = -k_1 u - mg \sin \theta.$$

Newton's law of motion relates this force sum to the tip-mass acceleration, so

$$\boxed{\sum F_x = m\ddot{u}(t) = -k_1 u - mg \sin \theta} \quad (\text{C.7.4})$$

The analysis assumes small transverse displacement and small slope, so $\sin \theta \approx \theta$. It also assumes a uniform Euler-Bernoulli beam, linear elastic behavior, negligible beam mass, a rigid tip mass, one horizontal degree of freedom, no damping, and no torsional or axial motion. Because the beam follows the equivalent static tip-load shape, $\theta = 3u/(2L)$. Substituting these two approximations into the force balance gives

$$m\ddot{u} + k_1 u + mg\theta = 0,$$

$$m\ddot{u} + \left(k_1 + \frac{3mg}{2L}\right) u = 0.$$

Thus, gravity supplies an additional linear restoring stiffness in this model.

$$\boxed{m\ddot{u} + \left(\frac{3EI_1}{L^3} + \frac{3mg}{2L}\right) u = 0} \quad (\text{C.7.5})$$

Since $m \neq 0$, division of the linearized equation by m identifies the coefficient of u with $\omega_{n,\text{FBP}}^2$. Therefore, the natural frequency of the flex-beam pendulum is

$$\omega_{n,\text{FBP}}^2 = \frac{k_1}{m} + \frac{3g}{2L} = \frac{3EI_1}{mL^3} + \frac{3g}{2L}.$$

Therefore,

$$\omega_{n,\text{FBP}} = \sqrt{\frac{3EI_1}{mL^3} + \frac{3g}{2L}} \quad (\text{C.7.6})$$

Using this definition, the equation of motion takes the normalized governing form

$$\ddot{u} + \omega_{n,\text{FBP}}^2 u = 0.$$

The equivalent stiffness is $k_{\text{eff}} = k_1 + 3mg/(2L)$, so the same equation can also be written as $m\ddot{u} + k_{\text{eff}}u = 0$.

$$\ddot{u} + \omega_{n,\text{FBP}}^2 u = 0, \quad k_{\text{eff}} = k_1 + \frac{3mg}{2L} \quad (\text{C.7.7})$$

Problem B.1 has the form $m\ddot{u} + ku = 0$ with natural frequency $\sqrt{k/m}$. The flex-beam pendulum has the same undamped single-degree-of-freedom form after replacing k by k_{eff} . The characteristic roots are $s = \pm i\omega_{n,\text{FBP}}$, so the general real solution is

$$u(t) = A \cos(\omega_{n,\text{FBP}}t) + B \sin(\omega_{n,\text{FBP}}t).$$

$$u(t) = A \cos(\omega_{n,\text{FBP}}t) + B \sin(\omega_{n,\text{FBP}}t) \quad (\text{C.7.8})$$

Writing the initial displacement and velocity as

$$u(0) = u_0, \quad \dot{u}(0) = \dot{u}_0.$$

Evaluating the general solution at $t = 0$ gives $A = u_0$. Differentiating the solution gives

$$\dot{u}(t) = -A\omega_{n,\text{FBP}} \sin(\omega_{n,\text{FBP}}t) + B\omega_{n,\text{FBP}} \cos(\omega_{n,\text{FBP}}t).$$

At $t = 0$, this becomes $\dot{u}_0 = B\omega_{n,\text{FBP}}$, so $B = \dot{u}_0/\omega_{n,\text{FBP}}$. Substitution gives the response for simultaneous initial displacement and initial velocity.

$$u(t) = u_0 \cos(\omega_{n,\text{FBP}}t) + \frac{\dot{u}_0}{\omega_{n,\text{FBP}}} \sin(\omega_{n,\text{FBP}}t) \quad (\text{C.7.9})$$

If flexure occurs about the major second moment I_2 , the elastic stiffness uses I_2 instead of I_1 . Therefore, the new stiffness and natural frequency are

$$k_2 = \frac{3EI_2}{L^3}, \quad \omega_{n,2} = \sqrt{\frac{3EI_2}{mL^3} + \frac{3g}{2L}}.$$

Since

$$\frac{I_2}{I_1} = \left(\frac{b}{h}\right)^2,$$

the major-axis elastic stiffness is larger by $(b/h)^2$. The gravity term is unchanged because it depends on m , g , and L , not on the bending axis. The corresponding initial-value response is obtained by replacing $\omega_{n,\text{FBP}}$ with $\omega_{n,2}$.

$$\boxed{\omega_{n,2} = \sqrt{\frac{3EI_2}{mL^3} + \frac{3g}{2L}}, \quad \frac{I_2}{I_1} = \left(\frac{b}{h}\right)^2} \quad (\text{C.7.10})$$

For $b > h$, $\omega_{n,2} > \omega_{n,\text{FBP}}$, so the major-axis motion oscillates faster and has a shorter period. In the limit that elastic stiffness dominates the gravity contribution, the frequency ratio approaches

$$\frac{\omega_{n,2}}{\omega_{n,\text{FBP}}} \rightarrow \sqrt{\frac{I_2}{I_1}} = \frac{b}{h}.$$

C.8 Calculate Flex-Beam Pendulum Response to Impact

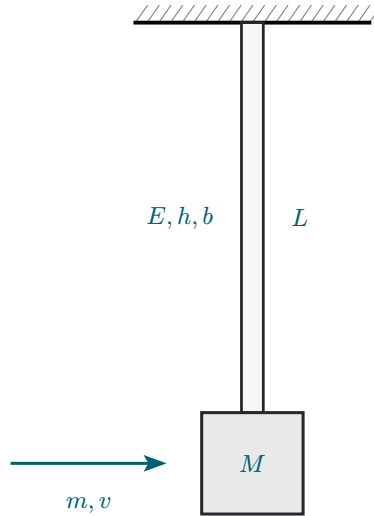


Figure 43: Impact excitation of a vertical flex-beam pendulum.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L = 568 \text{ mm}$ with a mass $M = 42 \text{ kg}$ attached at its end, as shown in Figure 43. The beam is steel with $E = 200 \text{ GPa}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$. The beam is initially at rest. A projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. The projectile orientation causes the beam to flex about its minor cross-sectional axis. The impact imparts an initial velocity to the system, which then oscillates.

Do the following.

- (1) Display input data.
- (2) Calculate the minor second moment of area I of the beam.
- (3) Calculate the minor cross-sectional stiffness EI .
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass.
- (6) Calculate the rad/s natural frequency $\omega_{n,\text{FBP}}$ and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it.
- (7) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (7).
- (9) Compare your results with those of Problem A.3.

Solution to C.8

Denote the projectile mass by m_p and the mass that oscillates after impact by $M_t = M + m_p$. The beam flexes about its minor axis, and the transverse displacement is measured from the vertical position. All calculations use SI units.

Quantity	Value	Unit
L	0.568	m
M	42	kg
m_p	0.018	kg
v_p	935	m/s
E	200×10^9	Pa
h	0.003	m
b	0.037	m
g	9.81	m/s ²

$$\boxed{M_t = M + m_p = 42.018 \text{ kg}} \quad (\text{C.8.1})$$

For the rectangular cross-section, the minor second moment of area is

$$I = \frac{bh^3}{12} = \frac{(0.037)(0.003)^3}{12} = 8.325 \times 10^{-11} \text{ m}^4 = 83.25 \text{ mm}^4.$$

$$\boxed{I = 8.325 \times 10^{-11} \text{ m}^4 = 83.25 \text{ mm}^4} \quad (\text{C.8.2})$$

Multiplying the second moment by the elastic modulus gives the flexural stiffness,

$$EI = (200 \times 10^9)(8.325 \times 10^{-11}) = 16.65 \text{ N m}^2.$$

$$\boxed{EI = 16.65 \text{ N m}^2} \quad (\text{C.8.3})$$

The tip deflection of a cantilever under a concentrated tip load is $\Delta = PL^3/(3EI)$. Therefore, the equivalent minor-axis spring constant is

$$k = \frac{P}{\Delta} = \frac{3EI}{L^3} = \frac{3(16.65)}{(0.568)^3} = 272.5778 \text{ N/m}.$$

$$\boxed{k = 272.58 \text{ N/m}} \quad (\text{C.8.4})$$

Because the collision is short compared with the vibration and the projectile embeds in the initially stationary mass, momentum is conserved.

$$m_p v_p + M(0) = (M + m_p) \dot{u}_0.$$

Solving for the common velocity immediately after impact gives

$$\dot{u}_0 = \frac{m_p v_p}{M + m_p} = \frac{(0.018)(935)}{42.018} = 0.4005426 \text{ m/s.}$$

$$\boxed{\dot{u}_0 = 0.40054 \text{ m/s}, \quad u_0 = 0} \quad (\text{C.8.5})$$

The initial displacement is zero because the beam is vertical at the instant of impact. The total mass M_t is then used in the flex-beam pendulum frequency from C.7. Thus,

$$\omega_{n,\text{FBP}} = \sqrt{\frac{k}{M_t} + \frac{3g}{2L}} = \sqrt{\frac{272.5778}{42.018} + \frac{3(9.81)}{2(0.568)}} = 5.69156 \text{ rad/s.}$$

The corresponding frequency and period are

$$f_{n,\text{FBP}} = \frac{\omega_{n,\text{FBP}}}{2\pi} = 0.905840 \text{ Hz}, \quad \tau_{\text{FBP}} = \frac{2\pi}{\omega_{n,\text{FBP}}} = 1.10395 \text{ s.}$$

$$\boxed{\omega_{n,\text{FBP}} = 5.69156 \text{ rad/s}, \quad f_{n,\text{FBP}} = 0.90584 \text{ Hz}, \quad \tau_{\text{FBP}} = 1.10395 \text{ s}} \quad (\text{C.8.6})$$

After impact, the system is an undamped flex-beam pendulum. Substituting $u_0 = 0$ and $\dot{u}(0) = \dot{u}_0$ into the initial-value solution gives

$$u(t) = \frac{\dot{u}_0}{\omega_{n,\text{FBP}}} \sin(\omega_{n,\text{FBP}} t).$$

The amplitude is

$$\frac{\dot{u}_0}{\omega_{n,\text{FBP}}} = 0.0703748 \text{ m} = 70.3748 \text{ mm.}$$

At $t_1 = 10 \text{ s}$, the response is

$$u(10) = \frac{0.4005426}{5.69156} \sin(5.69156(10)) = 0.0252474 \text{ m} = 25.2474 \text{ mm.}$$

$$\boxed{u(t) = 70.3748 \sin(5.69156t) \text{ mm}, \quad u(10 \text{ s}) = 25.2474 \text{ mm}} \quad (\text{C.8.7})$$

For ten oscillations, the plot extends through

$$t_{\text{end}} = N_{\text{osc}} \tau_{\text{FBP}} = 10(1.10395) = 11.0395 \text{ s.}$$

The first positive maximum occurs when $\sin(\omega_{n,\text{FBP}} t) = 1$, so

$$t_{\text{max}} = \frac{\pi}{2\omega_{n,\text{FBP}}} = 0.275987 \text{ s}, \quad u_{\text{max}} = \frac{\dot{u}_0}{\omega_{n,\text{FBP}}} = 70.3748 \text{ mm.}$$

The two markers on the plot identify the first maximum and the response at $t = 10 \text{ s}$.

C8. Response after an embedded-projectile impact

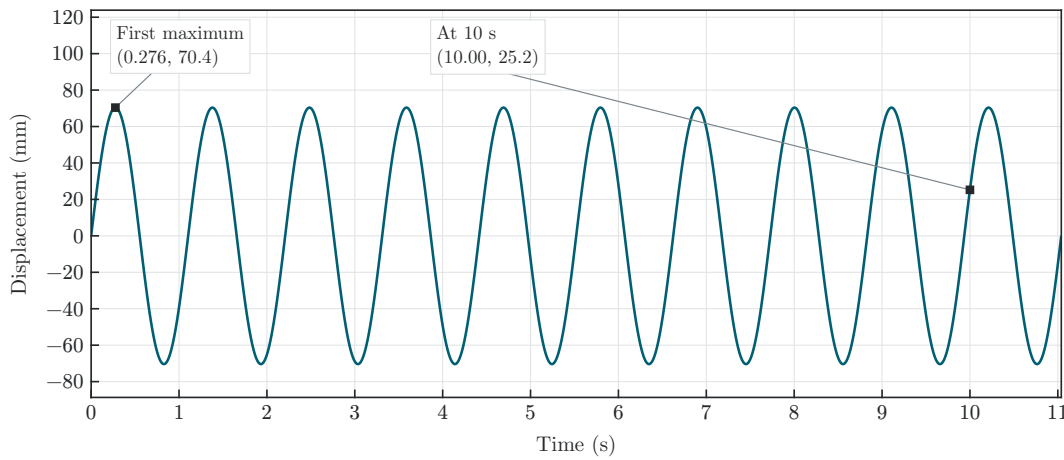


Figure 44: C.8. Ten-cycle response with data markers at the first maximum and at $t = 10 \text{ s}$.

$$N_{\text{osc}} = 10, \quad t_{\text{end}} = 11.0395 \text{ s}, \quad u_{\text{max}} = 70.3748 \text{ mm at } t = 0.27599 \text{ s} \quad (\text{C.8.8.a})$$

$$u(10 \text{ s}) = 25.2474 \text{ mm} \quad (\text{C.8.8.b})$$

MATLAB code. C.8 response plot.

```

01 L=.568; M=42; mp=.018; vp=935; E=200e9; h=.003; b=.037; g=9.81;
02 Nosc=10; t1=10;
03 I=b*h^3/12; k=3*E*I/L^3; Mt=M+mp;
04 v0=mp*vp/Mt; wn=sqrt(k/(Mt+3*g/(2*L))); tau=2*pi/wn;
05 t=linspace(0,Nosc*tau,2000); A=1000*v0/wn; u=A*sin(wn*t);
06 plot(t,u,'Color',[0 .427 .459],'LineWidth',1.5);
07 hold on; tmax=pi/(2*wn); plot(tmax,A,'ks');
08 plot(t1,A*sin(wn*t1),'ks'); grid on;
09 xlabel('Time (s)'); ylabel('Displacement (mm)');

```

Problem A.3 models a simple pendulum, while C.8 includes the cantilever elastic stiffness in addition to the gravity contribution. The impact velocities are nearly identical, but the C.8 frequency is higher because of the beam stiffness.

Quantity	C.8	A.3	Unit
$\dot{u}_0$	0.40054	0.39964	m/s
ω_n	5.69156	4.16687	rad/s
τ	1.10395	1.50789	s
$ u _{\max}$	70.3748	95.9085	mm
$u(10 \text{ s})$	25.2474	-70.6461	mm

The post-impact speeds differ by less than one percent. The flex-beam pendulum has a **36.6%** higher natural frequency and a **26.6%** smaller response amplitude than A.3. Its ten-second response is positive, whereas the A.3 response is negative because the two systems have different frequencies, periods, lengths, and impact data. The comparison shows the effect of the additional elastic restoring term, but it does not isolate that term because the input parameters also differ.

$$\boxed{\omega_{n,\text{FBP}} > \omega_{n,\text{A.3}}, \quad |u|_{\max,\text{FBP}} < |u|_{\max,\text{A.3}}} \quad (\text{C.8.9})$$

C.9 Complex Exponentials and Trigonometric Functions

- (1) Find the solutions of $z^4 = 1$ (the fourth roots of unity) and express them as:
 - (a) Complex numbers.
 - (b) Complex exponentials.
- (2) Write the complex number $e^{i\omega t}$ as a sum of trigonometric functions (hint, Euler's formula).
- (3) Calculate the phase difference ϕ in radians and degrees between the following signals:
 - (a) Compare $x_1(t) = \sin(\omega t)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (b) Compare $x_1(t) = \sin(\omega t + \pi/3)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (c) Compare $x_1(t) = \sin(\omega t + 45^\circ)$ and $x_2(t) = \sin(\omega t + 115^\circ)$.
- (4) What is the phase in radians and degrees of the following complex numbers:
 - (a) $e^{i\pi/2}$.
 - (b) $e^{i3\pi/2}$.
 - (c) $e^{i\pi/4}$.
 - (d) $e^{-i\pi/2}$.
 - (e) $e^{-i\pi/4}$.
- (5) Draw the locations on the unit circle of the complex numbers listed in item (4).
- (6) Calculate the magnitude and phase in degrees of the complex number $Z = 4 + 5i$.

Solution to C.9

Taking magnitudes in $z^4 = 1$ gives $|z| = 1$. Since every solution lies on the unit circle, it can be written as $z = e^{i\theta}$, which gives

$$e^{i4\theta} = 1 = e^{i2\pi k}, \quad k \in \mathbb{Z}.$$

Equating the exponents yields $4\theta = 2\pi k$, so $\theta = k\pi/2$. Since increasing k by four adds one full revolution, four consecutive values of k give all distinct roots.

$$\boxed{z \in \{1, i, -1, -i\}} \quad (\text{C.9.1.a})$$

Using $k = 0, 1, 2, 3$, the same roots in complex-exponential form are

$$\boxed{z_k = e^{ik\pi/2}, \quad k = 0, 1, 2, 3} \quad (\text{C.9.1.b})$$

Euler's identity combines the real and imaginary parts of a unit-magnitude complex exponential. Therefore, substituting $\alpha = \omega t$ gives

$$\boxed{e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)} \quad (\text{C.9.2})$$

Writing each sinusoid as $x(t) = \sin(\omega t + \phi)$ identifies ϕ as its phase. I define the phase difference as the phase of the second signal minus the phase of the first,

$$\Delta\phi = \phi_2 - \phi_1.$$

With this definition, a positive value means that $x_2(t)$ leads $x_1(t)$. For part (a), $\phi_1 = 0$ and $\phi_2 = 3\pi/4$, so

$$\boxed{\Delta\phi_a = \frac{3\pi}{4} = 135^\circ} \quad (\text{C.9.3.a})$$

For part (b), subtracting the first phase from the second and using a common denominator gives

$$\Delta\phi_b = \frac{3\pi}{4} - \frac{\pi}{3} = \frac{9\pi - 4\pi}{12} = \frac{5\pi}{12}.$$

Therefore,

$$\boxed{\Delta\phi_b = \frac{5\pi}{12} = 75^\circ} \quad (\text{C.9.3.b})$$

For part (c), subtracting the phases in degrees and then converting the result to radians gives

$$\Delta\phi_c = 115^\circ - 45^\circ = 70^\circ = 70 \left(\frac{\pi}{180} \right) = \frac{7\pi}{18}.$$

$$\boxed{\Delta\phi_c = \frac{7\pi}{18} = 70^\circ} \quad (\text{C.9.3.c})$$

The phase of $e^{i\theta}$ is θ modulo 2π . When an angle has more than one equivalent representation, I use the principal phase interval $(-\pi, \pi]$. Therefore, the requested phases are

$$\boxed{\arg(e^{i\pi/2}) = \frac{\pi}{2} = 90^\circ} \quad (\text{C.9.4.a})$$

For $e^{i3\pi/2}$, the given angle is 270° . Subtracting 360° gives the equivalent principal phase -90° ,

$$\boxed{\arg(e^{i3\pi/2}) = \frac{3\pi}{2} = 270^\circ \equiv -\frac{\pi}{2} = -90^\circ \text{ (principal)}} \quad (\text{C.9.4.b})$$

$$\boxed{\arg(e^{i\pi/4}) = \frac{\pi}{4} = 45^\circ} \quad (\text{C.9.4.c})$$

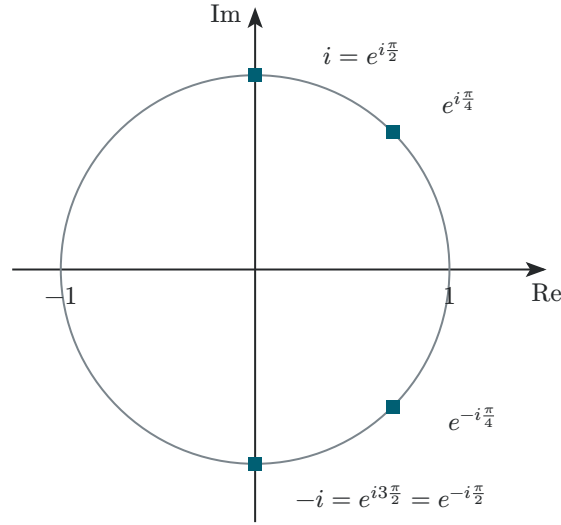
$$\boxed{\arg(e^{-i\pi/2}) = -\frac{\pi}{2} = -90^\circ} \quad (\text{C.9.4.d})$$

$$\boxed{\arg(e^{-i\pi/4}) = -\frac{\pi}{4} = -45^\circ} \quad (\text{C.9.4.e})$$

For item (5), the unit-circle coordinates are $(\cos \theta, \sin \theta)$. Therefore, $e^{i\pi/2}$ lies at $(0, 1)$, $e^{i3\pi/2}$ at $(0, -1)$, $e^{i\pi/4}$ at $(\sqrt{2}/2, \sqrt{2}/2)$, and $e^{-i\pi/4}$ at $(\sqrt{2}/2, -\sqrt{2}/2)$. Since the angles $3\pi/2$ and $-\pi/2$ differ by one full revolution, $e^{-i\pi/2}$ also lies at $(0, -1)$. Thus, the two expressions represent the same point,

$$e^{i3\pi/2} = e^{-i\pi/2} = -i.$$

Complex phases on the unit circle

**Figure 45:** C.9.5. Requested points on the unit circle.

$$e^{i3\pi/2} = e^{-i\pi/2} = -i \quad (\text{C.9.5})$$

For item (6), the real and imaginary parts of $Z = 4 + 5i$ are 4 and 5, respectively. Therefore, the magnitude is

$$|Z| = \sqrt{4^2 + 5^2} = \sqrt{41} = 6.4031.$$

Since Z lies in the first quadrant, the ordinary inverse tangent of the imaginary part divided by the real part gives its phase,

$$\arg Z = \tan^{-1} \left(\frac{5}{4} \right) = 0.8961 \text{ rad} = 51.3402^\circ.$$

$$|Z| = \sqrt{41} = 6.4031, \quad \arg Z = 0.8961 \text{ rad} = 51.3402^\circ \quad (\text{C.9.6})$$